

zip

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§ 1 Abstract

This paper proposes

- four views, `zip`, `zip_transform`, `adjacent`, and `adjacent_transform`,
- changes to `tuple` and `pair` necessary to make them usable as proxy references (necessary for `zip` and `adjacent`), and
- changes to `vector<bool>::reference` to make it usable as a proxy reference for writing,

all as described in section 3.2 of [\[P2214R0\]](#).

§ 2 Revision history

- R2: Incorporated LWG review feedback on 2021-05-21, 2021-05-28, 2021-06-04, and 2021-06-11. Account for integer-class types in the handling of `difference_type` and `size_type`. Rebasing to the expected post-2021-06 working draft.
- R1: Added feature test macro. Expanded discussion regarding 1) `operator==` for forward-or-weaker zip iterators and 2) `adjacent` on input ranges. Miscellaneous wording fixes (thanks to Barry Revzin and Tomasz Kamiński). Added a short example.

§ 3 Examples

```
std::vector v1 = {1, 2};
std::vector v2 = {'a', 'b', 'c'};
std::vector v3 = {3, 4, 5};

fmt::print("{}\n", std::views::zip(v1, v2));           // {(1,
               'a'), (2, 'b')}
fmt::print("{}\n", std::views::zip_transform(std::multiplies(), v1, v3)); // {3, 8}
fmt::print("{}\n", v2 | std::views::pairwise);         // {'a',
               'b'), ('b', 'c')}
fmt::print("{}\n", v3 | std::views::pairwise_transform(std::plus()));    // {7, 9}
```

§ 4 Discussion

The proposed wording below generally follows the design described in section 3.2 of [P2214R0], and the discussion in this paper assumes familiarity with that paper. This section focuses on deviations from and additions to that paper, as well as certain details in the design of the views that should be called out.

§ 4.1 Proxy reference changes

The basic rationale for changes to tuple and pair are described in exhaustive detail in [P2214R0] sections 3.2.1 and 3.2.2 and will not be repeated here. Several additions are worth noting:

- `common_type` and `basic_common_reference` specializations are added for tuple and pair. These are also required for tuple and pair to be usable as proxy references.
- `swap` for `const` tuple and `const` pair. Once tuples of references are made const-assignable, the default `std::swap` can be called for const tuples of references. However, that triple-move swap does the wrong thing:

```
int i = 1, j = 2;
const auto t1 = std::tie(i), t2 = std::tie(j);

// If std::swap(t1, t2); called the default triple-move std::swap then
// this would do
auto tmp = std::move(t1);
t1 = std::move(t2);
t2 = std::move(tmp);

// i == 2, j == 2
```

This paper therefore proposes adding overloads of `swap` for `const` tuples and pairs to correctly perform element-wise swap.

- Consistent with the scoped allocator protocol, allocator-extended constructors that correspond to the new tuple constructors have been added to tuple, and new overloads of `uses_allocator_construction_args` corresponding to the new pair constructors have been added as well.

§ 4.2 zip and zip_transform

§ 4.2.1 No common exposition-only base view

[P2214R0] proposes implementing `zip` and `zip_transform` to produce specializations of an exposition-only *iter-zip-transform-view*, which is roughly how they are implemented in `range-v3`. In the process of writing wording for these views, however, it has become apparent that the two views have enough differences that a common underlying view would need to have additional knobs to control the behavior (beyond the `value_type` issue already noted in the paper). The extra complexity required would likely negate any potential benefit from having a single underlying view.

Instead, the wording below specifies `zip_transform` largely in terms of `zip`. This significantly reduces specification duplication without sacrificing efficiency.

§ 4.2.2 tuple or pair?

In `range-v3`, zipping two views produced a range of pairs, while zipping any other number of ranges produce a range of tuples. This paper maintains that design for several reasons:

- Since `pair` is tuple-like, most common uses of the result (`get`, `apply`, structured bindings, etc.) would work just as well.
- Certain parts of the standard library, notably `map` and `friends`, are dependent on `pair`.
- `pair` implicitly converts to `tuple` if one is really needed, whereas constructing a `pair` from a `tuple` is more difficult.

§ 4.2.3 Zipping nothing?

As in `range-v3`, zipping nothing produces an `empty_view` of the appropriate type.

§ 4.2.4 When is zip_view a common_range?

A `common_range` is a range whose iterator and sentinel types are the same.

Obviously, when zipping a single range, the `zip_view` can be a `common_range` if the underlying range is.

When the `zip_view` is not bidirectional, it can be a `common_range` when every underlying view is a `common_range`. To handle differently-sized ranges, `iterator ==` is a logical OR: two iterators compare equal if one of the sub-iterators compare equal. Note that the domain of `==` only extends to iterators over the same underlying sequence; the use of logical OR is valid within that domain because the only valid operands to `==` are iterators obtained from incrementing `begin()` zero or more times and the iterator returned by `end()`.

When the `zip_view` is bidirectional (or stronger), however, it is now possible to iterate backwards from the end iterator (if it is indeed an iterator). As a result, we cannot simply construct the end iterator out of the end iterators of the views: if the views are different in size, iterating backwards from the end will give us elements that are not in the view at all (see [range-v3.1592]). Instead, we need to produce a “proper”

end iterator by advancing from `begin`; to be able compute end in constant time, we need all views to be random access and sized.

As end is only required to be amortized constant time, it is in theory possible to do a linear time traversal and cache the result. The additional benefit from such a design appears remote, and it has significant costs.

§ 4.2.4.1 Does that mean `views::enumerate(a_std_list)` isn't a `common_range`?

It's still an open question whether `enumerate` should be implemented in terms of `zip_view` or not. If it is specified in terms of `zip_view` (and produces a pair), as proposed in [P2214R0], it is easy to specify an separate `enumerate_view` that is implemented with `zip_view` but still produces a common range for this case.

§ 4.2.4.2 What about infinite ranges in general?

If `zip_view` can recognize when a range is infinite, then it is theoretically possible for it to be a `common_range` in the following two cases:

- Every range but one is an infinite random-access range, while the one finite range is a sized and common range (of any iterator strength). This is the general case of `enumerate` above.
- Every range is either random-access and sized, or random-access and infinite, and at least one range is sized.

The standard, however, generally does not recognize infinite ranges (despite providing `unreachable_sentinel`). It goes without saying that a complete design for infinite ranges support is outside the scope of this paper.

§ 4.2.5 `difference_type`

The `difference_type` of `zip_view` is the common type of the difference types of its constituent views. During LWG wording review, Casey Carter pointed out that we don't currently require integer-class types to have such a common type, or to be convertible to other integer-class types, which appears to be a defect in the specification of those types. This paper therefore adds wording to address the issue.

§ 4.3 `adjacent` and `adjacent_transform`

As `adjacent` is a specialized version of `zip`, most of the discussion in above applies, *mutatis mutandis*, to `adjacent` as well, and will not be repeated here.

§ 4.3.1 Naming

The wording below tentatively uses `adjacent` for the general functionality, and `pairwise` for the `N == 2` case. [P2214R0] section 3.2.5 suggests an alternative (`slide_as_tuple` for the general functionality and `adjacent` for the `N == 2` case). The author has a mild preference for the current names due to the somewhat unwieldiness of the name `slide_as_tuple`.

§ 4.3.2 Value type

The value type of `adjacent_view` is a homogeneous tuple or pair. Since array cannot hold references and is defined to be an aggregate, using it as the value type poses significant usability issues (even if we somehow get the `common_reference_with` requirements in `indirectly_readable` to work with even more tuple/pair changes).

§ 4.3.3 `common_range`

One notable difference from `zip` is that since `adjacent` comes from a single underlying view, it can be a `common_range` whenever its underlying view is.

§ 4.3.4 No input ranges

Because `adjacent` by definition holds multiple iterators to the same view, it requires forward ranges. It is true that the `N == 1` case could theoretically support input ranges, but that adds extra complexity and seems entirely pointless. Besides, someone desperate to wrap their input range in a single element tuple can just use `zip` instead.

During LEWG review of R0 of this paper it was suggested that `adjacent<N>` could support input views by caching the elements referred to by the last `N` iterators. Such a view would have significant differences from what is being proposed in this paper. For instance, because the reference obtained from an input iterator is invalidated on increment, the range will have to cache by value type, and so the reference type will have to be something like `tuple<range_value_t<V>&...>` (or perhaps even `tuple<range_value_t<V>...>&?`) instead of `tuple<range_reference_t<V>...>`. To be able to construct and update the cached values, the view would have to require underlying range's value type to be constructible and assignable from its reference type. And because we don't know what elements the user may desire to access, iterating through the view necessarily requires copying every element of the underlying view into the cache, which can be wasteful if not all elements need to be accessed. By comparison, iterating through the proposed `adjacent` copies exactly zero of the underlying range's elements.

Additionally, because input views provide much fewer operations and guarantees, they can often be implemented more efficiently than forward views. There has been an open range-v3 issue [[range-v3.704](#)] since 2017 (see also [this comment](#) from Eric Niebler on /r/cpp) to provide an API that downgrades a forward-or-stronger range to input for efficiency when the forward range's guarantees are not needed. Having a view adaptor that is significantly more expensive when given an input range would significantly damage the usability and teachability of such a design.

The author believes that the behavioral and performance characteristics of such a view is different enough from the `adjacent` proposed in this paper that it would be inappropriate to put them under the same name. It can be proposed separately if desired.

§ 4.3.5 `iter_swap`

Since the iterators of `adjacent_view` refer to potentially overlapping elements of the underlying view, `iter_swap` cannot really “exchange the values” of the range elements when the iterators overlap.

However, it does not appear to be possible to disable `ranges::iter_swap` (deleting or not providing `iter_swap` will simply fallback to the default implementation), and swapping non-overlapping iterators is still useful functionality. Thus, the wording below retains `iter_swap` but gives it a precondition that there is no overlap.

§ 5 Wording

This wording is relative to [\[N4885\]](#) after the application of [\[P2325R3\]](#), [\[LWG3526\]](#), and [\[LWG3527\]](#).

§ 5.1 tuple

1. Edit 20.5.2 [\[tuple.syn\]](#), header `<tuple>` synopsis, as indicated:

```
#include <compare>                                // see [compare.syn]

namespace std {
    // [tuple.tuple], class template tuple
    template<class... Types>
        class tuple;

+   template<class... TTypes, class... UTypes, template<class> class TQual,
+       template<class> class UQual>
+       requires requires { typename tuple<common_reference_t<TQual<TTypes>,
+           UQual<UTypes>>...>; }
+   struct basic_common_reference<tuple<TTypes...>, tuple<UTypes...>, TQual,
+       UQual> {
+       using type = tuple<common_reference_t<TQual<TTypes>,
+           UQual<UTypes>>...>;
+   };
+
+   template<class... TTypes, class... UTypes>
+       requires requires { typename tuple<common_type_t<TTypes, UTypes>...>; }
+   struct common_type<tuple<TTypes...>, tuple<UTypes...>> {
+       using type = tuple<common_type_t<TTypes, UTypes>...>;
+   };

    // [...]

    // [tuple.special], specialized algorithms
    template<class... Types>
        constexpr void swap(tuple<Types...>& x, tuple<Types...>& y)
            noexcept(see below);
+   template<class... Types>
+       constexpr void swap(const tuple<Types...>& x, const tuple<Types...>& y)
+           noexcept(see below);
}
```

2. Edit 20.5.3 [\[tuple.tuple\]](#), class template `tuple` synopsis, as indicated:

```
namespace std {
    template<class... Types>
```

```

class tuple {
public:
    // [tuple.cnstr], tuple construction
    constexpr explicit(see below) tuple();
    constexpr explicit(see below) tuple(const Types&...);           // only
        if sizeof...(Types) >= 1
    template<class... UTypes>
        constexpr explicit(see below) tuple(UTypes&&...);           // only
        if sizeof...(Types) >= 1

    tuple(const tuple&) = default;
    tuple(tuple&&) = default;

+   template<class... UTypes>
+   constexpr explicit(see below) tuple(tuple<UTypes...>&);
    template<class... UTypes>
        constexpr explicit(see below) tuple(const tuple<UTypes...>&);
    template<class... UTypes>
        constexpr explicit(see below) tuple(tuple<UTypes...>&&);
+   template<class... UTypes>
+   constexpr explicit(see below) tuple(const tuple<UTypes...>&&);

+   template<class U1, class U2>
+   constexpr explicit(see below) tuple(pair<U1, U2>&);           // only
        if sizeof...(Types) == 2
    template<class U1, class U2>
        constexpr explicit(see below) tuple(const pair<U1, U2>&);   // only
        if sizeof...(Types) == 2
    template<class U1, class U2>
        constexpr explicit(see below) tuple(pair<U1, U2>&&);           // only
        if sizeof...(Types) == 2
+   template<class U1, class U2>
+   constexpr explicit(see below) tuple(const pair<U1, U2>&&);   // only
        if sizeof...(Types) == 2

    // allocator-extended constructors
    template<class Alloc>
        constexpr explicit(see below)
            tuple(allocator_arg_t, const Alloc& a);
    template<class Alloc>
        constexpr explicit(see below)
            tuple(allocator_arg_t, const Alloc& a, const Types&...);
    template<class Alloc, class... UTypes>
        constexpr explicit(see below)
            tuple(allocator_arg_t, const Alloc& a, UTypes&&...);
    template<class Alloc>
        constexpr tuple(allocator_arg_t, const Alloc& a, const tuple&);
    template<class Alloc>
        constexpr tuple(allocator_arg_t, const Alloc& a, tuple&&);
+   template<class Alloc, class... UTypes>
+   constexpr explicit(see below)
+   tuple(allocator_arg_t, const Alloc& a, tuple<UTypes...>&);
    template<class Alloc, class... UTypes>
        constexpr explicit(see below)
            tuple(allocator_arg_t, const Alloc& a, const tuple<UTypes...>&);
    template<class Alloc, class... UTypes>

```



```

        constexpr explicit(see below)
        tuple(allocator_arg_t, const Alloc& a, tuple<UTypes...>&&);
+   template<class Alloc, class... UTypes>
+   constexpr explicit(see below)
+   tuple(allocator_arg_t, const Alloc& a, const tuple<UTypes...>&&);
+   template<class Alloc, class U1, class U2>
+   constexpr explicit(see below)
+   tuple(allocator_arg_t, const Alloc& a, pair<U1, U2>&&);
    template<class Alloc, class U1, class U2>
        constexpr explicit(see below)
        tuple(allocator_arg_t, const Alloc& a, const pair<U1, U2>&&);
    template<class Alloc, class U1, class U2>
        constexpr explicit(see below)
        tuple(allocator_arg_t, const Alloc& a, pair<U1, U2>&&);
+   template<class Alloc, class U1, class U2>
+   constexpr explicit(see below)
+   tuple(allocator_arg_t, const Alloc& a, const pair<U1, U2>&&);

    // [tuple.assign], tuple assignment
    constexpr tuple& operator=(const tuple&);
+   constexpr const tuple& operator=(const tuple&) const;
    constexpr tuple& operator=(tuple&&) noexcept(see below);
+   constexpr const tuple& operator=(tuple&&) const;

    template<class... UTypes>
        constexpr tuple& operator=(const tuple<UTypes...>&);
+   template<class... UTypes>
+   constexpr const tuple& operator=(const tuple<UTypes...>&) const;
    template<class... UTypes>
        constexpr tuple& operator=(tuple<UTypes...>&&);
+   template<class... UTypes>
+   constexpr const tuple& operator=(tuple<UTypes...>&&) const;

    template<class U1, class U2>
        constexpr tuple& operator=(const pair<U1, U2>&); // only if
        sizeof...(Types) == 2
+   template<class U1, class U2>
+   constexpr const tuple& operator=(const pair<U1, U2>&) const; //
        only if sizeof...(Types) == 2
    template<class U1, class U2>
        constexpr tuple& operator=(pair<U1, U2>&&); // only if
        sizeof...(Types) == 2
+   template<class U1, class U2>
+   constexpr const tuple& operator=(pair<U1, U2>&&) const; //
        only if sizeof...(Types) == 2

    // [tuple.swap], tuple swap
    constexpr void swap(tuple&) noexcept(see below);
+   constexpr void swap(const tuple&) const noexcept(see below);
};

// [...]
}

```

3. Edit 20.5.3.1 [\[tuple.cnstr\]](#) as indicated:

```

template<class... UTypes> constexpr explicit(see below)
    tuple(tuple<UTypes...>& u);
template<class... UTypes> constexpr explicit(see below) tuple(const
    tuple<UTypes...>& u);
template<class... UTypes> constexpr explicit(see below)
    tuple(tuple<UTypes...>&& u);
template<class... UTypes> constexpr explicit(see below) tuple(const
    tuple<UTypes...>&& u);

```

? Let I be the pack $0, 1, \dots, (\text{sizeof} \dots (\text{Types}) - 1)$. Let $\text{FWD}(u)$ be $\text{static_cast}<\text{decltype}(u)>(u)$.

? *Constraints:*

- (?.1) — $\text{sizeof} \dots (\text{Types})$ equals $\text{sizeof} \dots (\text{UTypes})$, and
- (?.2) — $(\text{is_constructible_v}<\text{Types}, \text{decltype}(\text{get}<I>(\text{FWD}(u)))> \ \&\& \ \dots)$ is true, and
- (?.3) — either $\text{sizeof} \dots (\text{Types})$ is not 1, or (when $\text{Types} \dots$ expands to T and $\text{UTypes} \dots$ expands to U) $\text{is_convertible_v}<\text{decltype}(u), T>$, $\text{is_constructible_v}<T, \text{decltype}(u)>$, and $\text{is_same_v}<T, U>$ are all false.

? *Effects:* For all i , initializes the i^{th} element of $*\text{this}$ with $\text{get}<i>(\text{FWD}(u))$.

? *Remarks:* The expression inside `explicit` is equivalent to: $!(\text{is_convertible_v}<\text{decltype}(\text{get}<I>(\text{FWD}(u))), \text{Types}> \ \&\& \ \dots)$

```

template<class... UTypes> constexpr explicit(see below) tuple(pair<U1, U2>&
    u);
template<class... UTypes> constexpr explicit(see below) tuple(const pair<U1,
    U2>& u);
template<class... UTypes> constexpr explicit(see below) tuple(pair<U1, U2>&&
    u);
template<class... UTypes> constexpr explicit(see below) tuple(const pair<U1,
    U2>&& u);

```

? Let $\text{FWD}(u)$ be $\text{static_cast}<\text{decltype}(u)>(u)$.

? *Constraints:*

- (?.1) — $\text{sizeof} \dots (\text{Types})$ is 2 and
- (?.2) — $\text{is_constructible_v}<T_0, \text{decltype}(\text{get}<0>(\text{FWD}(u)))>$ is true and
- (?.3) — $\text{is_constructible_v}<T_1, \text{decltype}(\text{get}<1>(\text{FWD}(u)))>$ is true.

? *Effects:* Initializes the first element with $\text{get}<0>(\text{FWD}(u))$ and the second element with $\text{get}<1>(\text{FWD}(u))$.

? *Remarks:* The expression inside `explicit` is equivalent to:
 $!(\text{is_convertible_v}<\text{decltype}(\text{get}<0>(\text{FWD}(u))), T_0> \ ||$
 $!(\text{is_convertible_v}<\text{decltype}(\text{get}<1>(\text{FWD}(u))), T_1>$

```
template<class... UTypes> constexpr explicit(see below) tuple(const-
    tuple<UTypes...>& u);
```

19 *Constraints:*

(19.1) — ~~sizeof...(Types) equals sizeof...(UTypes) and~~

(19.2) — ~~is_constructible_v<T_i, const U_i> is true for all *i*, and~~

(19.3) — ~~either sizeof...(Types) is not 1, or (when Types... expands to T and UTypes... expands to U) is_convertible_v<const tuple<U>&, T>, is_constructible_v<T, const tuple<U>&>, and is_same_v<T, U> are all false.~~

20 *Effects:* Initializes each element of ~~*this~~ with the corresponding element of ~~u~~.

21 *Remarks:* The expression inside ~~explicit~~ is equivalent to:

~~!conjunction_v<is_convertible<const UTypes&, Types>...>~~

```
template<class... UTypes> constexpr explicit(see below)
    tuple(tuple<UTypes...>&& u);
```

22 *Constraints:*

(22.1) — ~~sizeof...(Types) equals sizeof...(UTypes), and~~

(22.2) — ~~is_constructible_v<T_i, U_i> is true for all *i*, and~~

(22.3) — ~~either sizeof...(Types) is not 1, or (when Types... expands to T and UTypes... expands to U) is_convertible_v<tuple<U>, T>, is_constructible_v<T, tuple<U>>, and is_same_v<T, U> are all false.~~

23 *Effects:* For all *i*, initializes the *i*th element of ~~*this~~ with ~~std::forward<U_i>(get<i>(u))~~.

24 *Remarks:* The expression inside ~~explicit~~ is equivalent to:

~~!conjunction_v<is_convertible<UTypes, Types>...>~~

```
template<class U1, class U2> constexpr explicit(see below) tuple(const-
    pair<U1, U2>& u);
```

25 *Constraints:*

(25.1) — ~~sizeof...(Types) is 2,~~

(25.2) — ~~is_constructible_v<T₀, const U1&> is true, and~~

(25.3) — ~~is_constructible_v<T₁, const U2&> is true~~

26 *Effects:* Initializes the first element with ~~u.first~~ and the second element with ~~u.second~~.

27 ~~Remarks: The expression inside `explicit` is equivalent to:~~

~~`is_convertible_v<const U1&, T0> || is_convertible_v<const U2&, T1>`~~

~~`template<class U1, class U2> constexpr explicit(see below) tuple(pair<U1, U2>&& u);`~~

28 ~~Constraints:~~

(28.1) ~~— `sizeof...(Types)` is 2,~~

(28.2) ~~— `is_constructible_v<T0, U1>` is true, and~~

(28.3) ~~— `is_constructible_v<T1, U2>` is true~~

29 ~~Effects: Initializes the first element with `std::forward<U1>(u.first)` and the second element with `std::forward<U2>(u.second)`.~~

30 ~~Remarks: The expression inside `explicit` is equivalent to:~~

~~`is_convertible_v<U1, T0> || is_convertible_v<U2, T1>`~~

```
template<class Alloc>
constexpr explicit(see below)
tuple(allocator_arg_t, const Alloc& a);
template<class Alloc>
constexpr explicit(see below)
tuple(allocator_arg_t, const Alloc& a, const Types&...);
template<class Alloc, class... UTypes>
constexpr explicit(see below)
tuple(allocator_arg_t, const Alloc& a, UTypes&&...);
template<class Alloc>
constexpr tuple(allocator_arg_t, const Alloc& a, const
tuple&);
template<class Alloc>
constexpr tuple(allocator_arg_t, const Alloc& a, tuple&&);
+ template<class Alloc, class... UTypes>
+ constexpr explicit(see below)
+ tuple(allocator_arg_t, const Alloc& a, tuple<UTypes...>&);
template<class Alloc, class... UTypes>
constexpr explicit(see below)
tuple(allocator_arg_t, const Alloc& a, const
tuple<UTypes...>&);
template<class Alloc, class... UTypes>
constexpr explicit(see below)
tuple(allocator_arg_t, const Alloc& a, tuple<UTypes...>&&);
+ template<class Alloc, class... UTypes>
+ constexpr explicit(see below)
+ tuple(allocator_arg_t, const Alloc& a, const
tuple<UTypes...>&&);
+ template<class Alloc, class U1, class U2>
+ constexpr explicit(see below)
+ tuple(allocator_arg_t, const Alloc& a, pair<U1, U2>&);
template<class Alloc, class U1, class U2>
constexpr explicit(see below)
tuple(allocator_arg_t, const Alloc& a, const pair<U1, U2>&);
```

```

    template<class Alloc, class U1, class U2>
    constexpr explicit(see below)
    tuple(allocator_arg_t, const Alloc& a, pair<U1, U2>&&);
+ template<class Alloc, class U1, class U2>
+   constexpr explicit(see below)
+   tuple(allocator_arg_t, const Alloc& a, const pair<U1,
+       U2>&&);

```

31 *Preconditions:* Alloc meets the *Cpp17Allocator* requirements (Table 38).

32 *Effects:* Equivalent to the preceding constructors except that each element is constructed with uses-allocator construction.

4. Add the following to 20.5.3.2 [\[tuple.assign\]](#):

```
constexpr const tuple& operator=(const tuple& u) const;
```

? *Constraints:* (is_copy_assignable_v<const Types> && ...) is true.

? *Effects:* Assigns each element of u to the corresponding element of *this.

? *Returns:* *this.

```
constexpr const tuple& operator=(tuple&& u) const;
```

? *Constraints:* (is_assignable_v<const Types&, Types> && ...) is true.

? *Effects:* For all *i*, assigns std::forward<*T_iii*

? *Returns:* *this.

```
template<class... UTypes> constexpr const tuple& operator=(const
    tuple<UTypes...>& u) const;
```

? *Constraints:*

(?.1) — sizeof...(Types) equals sizeof...(UTypes) and

(?.2) — (is_assignable_v<const Types&, const UTypes&> && ...) is true.

? *Effects:* Assigns each element of u to the corresponding element of *this.

? *Returns:* *this.

```
template<class... UTypes> constexpr const tuple& operator=
    (tuple<UTypes...>&& u) const;
```

? *Constraints:*

(?.1) — sizeof...(Types) equals sizeof...(UTypes) and

(?.2) — (is_assignable_v<const Types&, UTypes> && ...) is true.

? *Effects:* For all *i*, assigns std::forward<*U_iii*

? *Returns:* `*this`.

```
template<class U1, class U2> constexpr const tuple& operator=(const pair<U1,
    U2>& u) const;
```

? *Constraints:*

(?.1) — `sizeof...(Types)` is 2,

(?.2) — `is_assignable_v<const T0&, const U1&>` is true, and

(?.3) — `is_assignable_v<const T1&, const U2&>` is true

? *Effects:* Assigns `u.first` to the first element and `u.second` to the second element.

? *Returns:* `*this`.

```
template<class U1, class U2> constexpr const tuple& operator=(pair<U1, U2>&&
    u) const;
```

? *Constraints:*

(?.1) — `sizeof...(Types)` is 2,

(?.2) — `is_assignable_v<const T0&, U1>` is true, and

(?.3) — `is_assignable_v<const T1&, U2>` is true

? *Effects:* Assigns `std::forward<U1>(u.first)` to the first element and `std::forward<U2>(u.second)` to the second element.

? *Returns:* `*this`.

5. Edit 20.5.3.3 [\[tuple.swap\]](#) as indicated:

```
constexpr void swap(tuple& rhs) noexcept(see below);
+ constexpr void swap(const tuple& rhs) const noexcept(see below);
```

? *Mandates:*

(?.1) — For the first overload, `(is_swappable_v<Types> && ...)` is true.

(?.2) — For the second overload, `(is_swappable_v<const Types> && ...)` is true.

1 *Preconditions:* Each element in `*this` is swappable with (16.4.4.3 [\[swappable.requirements\]](#)) the corresponding element in `rhs`.

2 *Effects:* Calls `swap` for each element in `*this` and its corresponding element in `rhs`.

3 *Throws:* Nothing unless one of the element-wise swap calls throws an exception.

4 *Remarks:* The expression inside `noexcept` is equivalent to the logical AND of the following expressions: `is_nothrow_swappable_v<Ti>`, where T_i is the *i*th type in `Types`.

(4.1) — `(is_nothrow_swappable_v<Types> && ...)` for the first overload.

(4.2) — `(is_nothrow_swappable_v<const Types> && ...)` for the second overload.

6. Edit 20.5.10 [\[tuple.special\]](#) as indicated:

```
template<class... Types>
constexpr void swap(tuple<Types...>& x, tuple<Types...>& y)
    noexcept(see below);
+ template<class... Types>
+ constexpr void swap(const tuple<Types...>& x, const
    tuple<Types...>& y) noexcept(see below);
```

1 *Constraints:* `is_swappable_v<T>` is true for every type T in `Types`.

(1.1) — For the first overload, `(is_swappable_v<Types> && ...)` is true.

(1.2) — For the second overload, `(is_swappable_v<const Types> && ...)` is true.

2 *Effects:* As if by `x.swap(y)`.

3 *Remarks:* The expression inside `noexcept` is equivalent to:
`noexcept(x.swap(y))`.

§ 5.2 pair

1. Edit 20.2.1 [\[utility.syn\]](#), header `<utility>` synopsis, as indicated:

```
#include <compare>           // see [compare.syn]
#include <initializer_list>   // see [initializer.list.syn]

namespace std {
    // [...]

    // [pairs], class template pair
    template<class T1, class T2>
        struct pair;

+   template<class T1, class T2, class U1, class U2, template<class> class
        TQual, template<class> class UQual>
+   requires requires { typename pair<common_reference_t<TQual<T1>,
        UQual<U1>>,
+   common_reference_t<TQual<T2>,
        UQual<U2>>>; }
+   struct basic_common_reference<pair<T1, T2>, pair<U1, U2>, TQual, UQual> {
+   using type = pair<common_reference_t<TQual<T1>, UQual<U1>>,
+   common_reference_t<TQual<T2>, UQual<U2>>>;
```

```

+ };
+
+ template<class T1, class T2, class U1, class U2>
+ requires requires { typename pair<common_type_t<T1, U1>,
+   common_type_t<T2, U2>>; }
+ struct common_type<pair<T1, T2>, pair<U1, U2>> {
+   using type = pair<common_type_t<T1, U1>, common_type_t<T2, U2>>;
+ };

// [pairs.spec], pair specialized algorithms
template<class T1, class T2>
constexpr bool operator==(const pair<T1, T2>&, const pair<T1, T2>&);
template<class T1, class T2>
constexpr common_comparison_category_t<synth-three-way-result<T1>,
                                         synth-three-way-result<T2>>
operator<=>(const pair<T1, T2>&, const pair<T1, T2>&);

template<class T1, class T2>
constexpr void swap(pair<T1, T2>& x, pair<T1, T2>& y)
noexcept(noexcept(x.swap(y)));
+ template<class T1, class T2>
+ constexpr void swap(const pair<T1, T2>& x, const pair<T1, T2>& y)
+   noexcept(noexcept(x.swap(y)));
}

```

2. Edit 20.4.2 [\[pairs.pair\]](#) as indicated:

```

namespace std {
  template<class T1, class T2>
  struct pair {
    using first_type = T1;
    using second_type = T2;

    T1 first;
    T2 second;

    pair(const pair&) = default;
    pair(pair&&) = default;
    constexpr explicit(see below) pair();
    constexpr explicit(see below) pair(const T1& x, const T2& y);
    template<class U1, class U2>
      constexpr explicit(see below) pair(U1&& x, U2&& y);
+   template<class U1, class U2>
+     constexpr explicit(see below) pair(pair<U1, U2>& p);
    template<class U1, class U2>
      constexpr explicit(see below) pair(const pair<U1, U2>& p);
    template<class U1, class U2>
      constexpr explicit(see below) pair(pair<U1, U2>&& p);
+   template<class U1, class U2>
+     constexpr explicit(see below) pair(const pair<U1, U2>&& p);
    template<class... Args1, class... Args2>
      constexpr pair(piecewise_construct_t,
                     tuple<Args1...> first_args, tuple<Args2...>
                     second_args);

    constexpr pair& operator=(const pair& p);

```



```

+ constexpr const pair& operator=(const pair& p) const;
  template<class U1, class U2>
    constexpr pair& operator=(const pair<U1, U2>& p);
+ template<class U1, class U2>
+   constexpr const pair& operator=(const pair<U1, U2>& p) const;
  constexpr pair& operator=(pair&& p) noexcept(see below);
+ constexpr const pair& operator=(pair&& p) const;
  template<class U1, class U2>
    constexpr pair& operator=(pair<U1, U2>&& p);
+ template<class U1, class U2>
+   constexpr const pair& operator=(pair<U1, U2>&& p) const;

  constexpr void swap(pair& p) noexcept(see below);
+ constexpr void swap(const pair& p) const noexcept(see below);
};

template<class T1, class T2>
  pair(T1, T2) -> pair<T1, T2>;
}

```

[...]

```

template<class U1, class U2> constexpr explicit(see below) pair(pair<U1,
    U2>& p);
template<class U1, class U2> constexpr explicit(see below) pair(const
    pair<U1, U2>& p);
template<class U1, class U2> constexpr explicit(see below) pair(pair<U1,
    U2>&& p);
template<class U1, class U2> constexpr explicit(see below) pair(const
    pair<U1, U2>&& p);

```

? Let FWD(u) be `static_cast<decltype(u)>(u)`.

? *Constraints:*

(?.1) — `is_constructible_v<first_type, decltype(get<0>(FWD(p)))>` is true and

(?.2) — `is_constructible_v<second_type, decltype(get<1>(FWD(p)))>` is true.

? *Effects:* Initializes first with `get<0>(FWD(p))` and second with `get<1>(FWD(p))`.

? *Remarks:* The expression inside `explicit` is equivalent to:

```

!is_convertible_v<decltype(get<0>(FWD(p))), first_type> ||
!is_convertible_v<decltype(get<1>(FWD(p))), second_type>.

```

```

template<class U1, class U2> constexpr explicit(see below) pair(const
    pair<U1, U2>& p);

```

14 ~~*Constraints:*~~

(14.1) — ~~`is_constructible_v<first_type, const U1&>` is true and~~

(14.2) — ~~is_constructible_v<second_type, const U2> is true.~~

15 ~~Effects: Initializes members from the corresponding members of the argument.~~

16 ~~Remarks: The expression inside explicit is equivalent to:~~
~~is_convertible_v<const U1&, first_type> ||~~
~~is_convertible_v<const U2&, second_type>~~

```
template<class U1, class U2> constexpr explicit(see below) pair(pair<U1,
    U2>&& p);
```

17 ~~Constraints:~~

(17.1) — ~~is_constructible_v<first_type, U1> is true and~~

(17.2) — ~~is_constructible_v<second_type, U2> is true.~~

18 ~~Effects: Initializes first with std::forward<U1>(p.first) and second with~~
~~std::forward<U2>(p.second).~~

19 ~~Remarks: The expression inside explicit is equivalent to:~~
~~is_convertible_v<U1, first_type> || is_convertible_v<U2,~~
~~second_type>~~

```
constexpr const pair& operator=(const pair& p) const;
```

? *Constraints:*

(?.1) — is_copy_assignable<const first_type> is true and

(?.2) — is_copy_assignable<const second_type> is true.

? *Effects:* Assigns p.first to first and p.second to second.

? *Returns:* *this.

```
template<class U1, class U2> constexpr const pair& operator=(const pair<U1,
    U2>& p) const;
```

? *Constraints:*

(?.1) — is_assignable_v<const first_type&, const U1&> is true, and

(?.2) — is_assignable_v<const second_type&, const U2&> is true.

? *Effects:* Assigns p.first to first and p.second to second.

? *Returns:* *this.

```
constexpr const pair& operator=(pair&& p) const;
```

? *Constraints:*

(?.1) — is_assignable<const first_type&, first_type> is true and

(?.2) — `is_assignable<const second_type&, second_type>` is true.

? *Effects:* Assigns `std::forward<first_type>(p.first)` to `first` and `std::forward<second_type>(p.second)` to `second`.

? *Returns:* `*this`.

```
template<class U1, class U2> constexpr const pair& operator=(pair<U1, U2>&&
    p) const;
```

? *Constraints:*

(?.1) — `is_assignable_v<const first_type&, U1>` is true, and

(?.2) — `is_assignable_v<const second_type&, U2>` is true.

? *Effects:* Assigns `std::forward<U1>(p.first)` to `first` and `std::forward<U2>(u.second)` to `second`.

? *Returns:* `*this`.

```
constexpr void swap(pair& p) noexcept(see below);
+ constexpr void swap(const pair& p) const noexcept(see below);
```

? *Mandates:*

(?.1) — For the first overload, `is_swappable_v<T1>` is true and `is_swappable_v<T2>` is true.

(?.2) — For the second overload, `is_swappable_v<const T1>` is true and `is_swappable_v<const T2>` is true.

35 *Preconditions:* `first` is swappable with (16.4.4.3 [\[swappable.requirements\]](#)) `p.first` and `second` is swappable with `p.second`.

36 *Effects:* Swaps `first` with `p.first` and `second` with `p.second`.

37 *Remarks:* The expression inside `noexcept` is equivalent to

(37.1) — `is_nothrow_swappable_v<first_type> && is_nothrow_swappable_v<second_type>` for the first overload.

(37.2) — `is_nothrow_swappable_v<const first_type> && is_nothrow_swappable_v<const second_type>` for the second overload.

3. Edit 20.4.3 [\[pairs.spec\]](#) as indicated:

```
template<class T1, class T2>
    constexpr void swap(pair<T1, T2>& x, pair<T1, T2>& y)
        noexcept(noexcept(x.swap(y)));
+ template<class T1, class T2>
+     constexpr void swap(const pair<T1, T2>& x, const pair<T1, T2>&
        y) noexcept(noexcept(x.swap(y)));
```

3 Constraints:

- (3.1) — For the first overload, `is_swappable_v<T1>` is `true` and `is_swappable_v<T2>` is `true`.
- (3.2) — For the second overload, `is_swappable_v<const T1>` is `true` and `is_swappable_v<const T2>` is `true`.

⁴ *Effects*: Equivalent to `x.swap(y)`.

4. Edit 20.10.2 [[memory.syn](#)], header `<memory>` synopsis, as indicated:

[illegible]

```
// [...]  
}
```

5. Edit 20.10.8.2 [\[allocator.uses.construction\]](#) as indicated:

```
+ template<class T, class Alloc, class U, class V>  
+ constexpr auto uses_allocator_construction_args(const Alloc&  
+ alloc,  
+ pair<U,V>& pr)  
+ noexcept;  
template<class T, class Alloc, class U, class V>  
constexpr auto uses_allocator_construction_args(const Alloc&  
alloc,  
pair<U,V>& pr) noexcept; const
```

12 *Constraints*: T is a specialization of pair.

13 *Effects*: Equivalent to:

```
return uses_allocator_construction_args<T>(alloc,  
piecewise_construct,  
forward_as_tuple(pr.first),  
forward_as_tuple(pr.second));
```

```
template<class T, class Alloc, class U, class V>  
constexpr auto uses_allocator_construction_args(const Alloc&  
alloc,  
pair<U,V>&&  
pr) noexcept;
```

```
+ template<class T, class Alloc, class U, class V>  
+ constexpr auto uses_allocator_construction_args(const Alloc&  
+ alloc,  
+ pair<U,V>&& pr) noexcept; const
```

14 *Constraints*: T is a specialization of pair.

15 *Effects*: Equivalent to:

```
return uses_allocator_construction_args<T>(alloc,  
piecewise_construct,  
forward_as_tuple(get<0>(std::move(pr))),  
forward_as_tuple(get<1>(std::move(pr))));
```

§ 5.3 vector<bool>::reference

Edit 22.3.12 [\[vector.bool\]](#), class template partial specialization vector<bool, Allocator> synopsis, as indicated:

```

namespace std {
    template<class Allocator>
    class vector<bool, Allocator> {
    public:
        // [...]

        // bit reference
        class reference {
            friend class vector;
            constexpr reference() noexcept;
        public:
            constexpr reference(const reference&) = default;
            constexpr ~reference();
            constexpr operator bool() const noexcept;
            constexpr reference& operator=(const bool x) noexcept;
            constexpr reference& operator=(const reference& x) noexcept;
+         constexpr const reference& operator=(bool x) const noexcept;
            constexpr void flip() noexcept;    // flips the bit
        };

        // [...]
    };
}

```

§ 5.4 Conversion and common type for integer-class types

Edit 23.3.4.4 [\[iterator.concept.winc\]](#) as indicated:

- 6 Expressions of integer-class type are explicitly convertible to any integral type **as well as any integer-class type**. Expressions of integral type are both implicitly and explicitly convertible to any integer-class type. Conversions between integral and integer-class types **and between two integer-class types** do not exit via an exception.

[...]

- 11 A type *I* other than cv `bool` is *integer-like* if it models `integral<I>` or if it is an integer-class type. An integer-like type *I* is *signed-integer-like* if it models `signed_integral<I>` or if it is a signed-integer-class type. An integer-like type *I* is *unsigned-integer-like* if it models `unsigned_integral<I>` or if it is an unsigned-integer-class type.

- ? For any two integer-like types *I1* and *I2*, at least one of which is an integer-class type, `common_type_t<I1, I2>` denotes an integer-like type whose width is not less than that of *I1* or *I2*. If both *I1* and *I2* are signed-integer-like types, then `common_type_t<I1, I2>` is also a signed-integer-like type.

§ 5.5 Addition to `<ranges>`

Add the following to 24.2 [\[ranges.syn\]](#), header `<ranges>` synopsis:

```

// [...]
namespace std::ranges {

```

```

// [...]

// [range.zip], zip view
template<input_range... Views>
    requires (view<Views> && ...) && (sizeof...(Views) > 0)
class zip_view;

template<class... Views>
    inline constexpr bool enable_borrowed_range<zip_view<Views...>> =
        (enable_borrowed_range<Views> && ...);

namespace views { inline constexpr unspecified zip = unspecified; }

// [range.zip.transform], zip transform view
template<copy_constructible F, input_range... Views>
    requires (view<Views> && ...) && (sizeof...(Views) > 0) && is_object_v<F> &&
        regular_invocable<F&, range_reference_t<Views>...> &&
        can-reference<invoke_result_t<F&, range_reference_t<Views>...>>
class zip_transform_view;

namespace views { inline constexpr unspecified zip_transform = unspecified; }

// [range.adjacent], adjacent view
template<forward_range V, size_t N>
    requires view<V> && (N > 0)
class adjacent_view;

template<class V, size_t N>
    inline constexpr bool enable_borrowed_range<adjacent_view<V, N>> =
        enable_borrowed_range<V>;

namespace views {
    template<size_t N>
        inline constexpr unspecified adjacent = unspecified;
        inline constexpr auto pairwise = adjacent<2>;
}

// [range.adjacent.transform], adjacent transform view
template<forward_range V, copy_constructible F, size_t N>
    requires see below
class adjacent_transform_view;

namespace views {
    template<size_t N>
        inline constexpr unspecified adjacent_transform = unspecified;
        inline constexpr auto pairwise_transform = adjacent_transform<2>;
}
}

```

§ 5.6 zip

Add the following subclause to 24.7 [\[range.adaptors\]](#).

§ 24.7.? Zip view [\[range.zip\]](#)

§ 24.7.?.1 Overview [range.zip.overview]

- 1 zip_view takes any number of views and produces a view of tuples of references to the corresponding elements of the constituent views.
- 2 The name views::zip denotes a customization point object (16.3.3.3.6 [\[customization.point.object\]](#)). Given a pack of subexpressions Es..., the expression views::zip(Es...) is expression-equivalent to
 - .1) — decay-copy(views::empty<tuple<>>) if Es is an empty pack,
 - .2) — otherwise, zip_view<views::all_t<decltype((Es))>...>(Es...).

[Example 1:

```
vector v = {1, 2};
list l = {'a', 'b', 'c'};

auto z = views::zip(v, l);
range_reference_t<decltype(z)> f = z.front();    // f is a pair<int&, char&> that
                                                // refers to the first element of v and l

for (auto&& [x, y] : z) {
    cout << '(' << x << ", " << y << " "; // prints: (1, a) (2, b)
}
```

— end example]

§ 24.7.?.2 Class template zip_view [range.zip.view]

```
namespace std::ranges {

template<class... Rs>
concept zip-is-common =                // exposition only
    (sizeof...(Rs) == 1 && (common_range<Rs> && ...)) ||
    (!(bidirectional_range<Rs> && ...) && (common_range<Rs> && ...)) ||
    ((random_access_range<Rs> && ...) && (sized_range<Rs> && ...));

template<class... Ts>
using tuple-or-pair = see below;    // exposition only

template<class F, class Tuple>
constexpr auto tuple-transform(F&& f, Tuple&& tuple) // exposition only
{
    return apply([&]class... Ts(Ts&&... elements){
        return tuple-or-pair<invoke_result_t<F&, Ts>...>(
            invoke(f, std::forward<Ts>(elements))...
        );
    }, std::forward<Tuple>(tuple));
}

template<class F, class Tuple>
constexpr void tuple-for-each(F&& f, Tuple&& tuple) // exposition only
{
    apply([&]class... Ts(Ts&&... elements){
        (invoke(f, std::forward<Ts>(elements)), ...);
    }, tuple);
}
```



```

}, std::forward<Tuple>(tuple));
}

template<input_range... Views>
requires (view<Views> && ...) && (sizeof...(Views) > 0)
class zip_view : public view_interface<zip_view<Views...>>{
    tuple<Views...> views_;           // exposition only

    template<bool> class iterator;    // exposition only
    template<bool> class sentinel;    // exposition only

public:
    zip_view() = default;
    constexpr explicit zip_view(Views... views);

    constexpr auto begin() requires (!(simple-view<Views> && ...)) {
        return iterator<false>(tuple-transform(ranges::begin, views_));
    }
    constexpr auto begin() const requires (range<const Views> && ...) {
        return iterator<true>(tuple-transform(ranges::begin, views_));
    }

    constexpr auto end() requires (!(simple-view<Views> && ...)) {
        if constexpr (!zip-is-common<Views...>) {
            return sentinel<false>(tuple-transform(ranges::end, views_));
        }
        else if constexpr ((random_access_range<Views> && ...)) {
            return begin() + iter_difference_t<iterator<false>>(size());
        }
        else {
            return iterator<false>(tuple-transform(ranges::end, views_));
        }
    }

    constexpr auto end() const requires (range<const Views> && ...) {
        if constexpr (!zip-is-common<const Views...>) {
            return sentinel<true>(tuple-transform(ranges::end, views_));
        }
        else if constexpr ((random_access_range<const Views> && ...)) {
            return begin() + iter_difference_t<iterator<true>>(size());
        }
        else {
            return iterator<true>(tuple-transform(ranges::end, views_));
        }
    }

    constexpr auto size() requires (sized_range<Views> && ...);
    constexpr auto size() const requires (sized_range<const Views> && ...);
};

template<class... Rs>
zip_view(Rs&&...) -> zip_view<views::all_t<Rs>...>;
}

```

- 1 Given some pack of types `Ts`, the alias template `tuple-or-pair` is defined as follows:
 - .1) — If `sizeof...(Ts)` is 2, `tuple-or-pair<Ts...>` denotes `pair<Ts...>`.
 - .2) — Otherwise, `tuple-or-pair<Ts...>` denotes `tuple<Ts...>`.
- 2 Two `zip_view` objects have the same underlying sequence if and only if the corresponding elements of `views_` are equal (18.2 [\[concepts.equality\]](#)) and have the same underlying sequence. [*Note 1*: In particular, comparison of iterators obtained from `zip_view` objects that do not have the same underlying sequence is not required to produce meaningful results (23.3.4.11 [\[iterator.concept.forward\]](#)). — *end note*]

```
constexpr explicit zip_view(Views... views);
```

- 3 *Effects*: Initializes `views_` with `std::move(views)....`

```
constexpr auto size() requires (sized_range<Views> && ...);
constexpr auto size() const requires (sized_range<const Views> && ...);
```

- 4 *Effects*: Equivalent to:

```
return apply([](auto... sizes){
    using CT = make_unsigned_like_t<common_type_t<decltype(sizes)...>>;
    return ranges::min({CT(sizes)...});
}, tuple-transform(ranges::size, views_));
```

§ 24.7.?.3 Class template `zip_view::iterator` [\[range.zip.iterator\]](#)

```
namespace std::ranges {
    template<bool Const, class... Views>
        concept all-random-access = (random_access_range<maybe-const<Const, Views>> &&
            ...); // exposition only
    template<bool Const, class... Views>
        concept all-bidirectional = (bidirectional_range<maybe-const<Const, Views>> &&
            ...); // exposition only
    template<bool Const, class... Views>
        concept all-forward = (forward_range<maybe-const<Const, Views>> && ...);
        // exposition only

    template<input_range... Views>
        requires (view<Views> && ...) && (sizeof...(Views) > 0)
    template<bool Const>
    class zip_view<Views...>::iterator {
        tuple-or-pair<iterator_t<maybe-const<Const, Views>>...> current_;
        // exposition only
        constexpr explicit iterator(tuple-or-pair<iterator_t<maybe-const<Const,
            Views>>...> current); // exposition only
    public:
        using iterator_category = input_iterator_tag; // not always present
        using iterator_concept = see below;
        using value_type = tuple-or-pair<range_value_t<maybe-const<Const, Views>>...>;
        using difference_type = common_type_t<range_difference_t<maybe-const<Const,
            Views>>...>;
```

```

iterator() = default;
constexpr iterator(iterator<!Const> i)
    requires Const && (convertible_to<iterator_t<Views>, iterator_t<maybe-
        const<Const, Views>>> && ...);

constexpr auto operator*() const;
constexpr iterator& operator++();
constexpr void operator++(int);
constexpr iterator operator++(int) requires all-forward<Const, Views...>;

constexpr iterator& operator--() requires all-bidirectional<Const, Views...>;
constexpr iterator operator--(int) requires all-bidirectional<Const, Views...>;

constexpr iterator& operator+=(difference_type x)
    requires all-random-access<Const, Views...>;
constexpr iterator& operator-=(difference_type x)
    requires all-random-access<Const, Views...>;

constexpr auto operator[](difference_type n) const
    requires all-random-access<Const, Views...>;

friend constexpr bool operator==(const iterator& x, const iterator& y)
    requires (equality_comparable<iterator_t<maybe-const<Const, Views>>> && ...);

friend constexpr bool operator<(const iterator& x, const iterator& y)
    requires all-random-access<Const, Views...>;
friend constexpr bool operator>(const iterator& x, const iterator& y)
    requires all-random-access<Const, Views...>;
friend constexpr bool operator<=(const iterator& x, const iterator& y)
    requires all-random-access<Const, Views...>;
friend constexpr bool operator>=(const iterator& x, const iterator& y)
    requires all-random-access<Const, Views...>;
friend constexpr auto operator<=>(const iterator& x, const iterator& y)
    requires all-random-access<Const, Views...> &&
        (three_way_comparable<iterator_t<maybe-const<Const, Views>>> && ...);

friend constexpr iterator operator+(const iterator& i, difference_type n)
    requires all-random-access<Const, Views...>;
friend constexpr iterator operator+(difference_type n, const iterator& i)
    requires all-random-access<Const, Views...>;
friend constexpr iterator operator-(const iterator& i, difference_type n)
    requires all-random-access<Const, Views...>;
friend constexpr difference_type operator-(const iterator& x, const iterator& y)
    requires (sized_sentinel_for<iterator_t<maybe-const<Const, Views>>,
        iterator_t<maybe-const<Const, Views>>> && ...);

friend constexpr auto iter_move(const iterator& i)
    noexcept(see below);

friend constexpr void iter_swap(const iterator& l, const iterator& r)
    noexcept(see below)
    requires (indirectly_swappable<iterator_t<maybe-const<Const, Views>>> && ...);
};
}

```

¹ `iterator::iterator_concept` is defined as follows:

- .1) — If *all-random-access*<Const, Views...> is modeled, then `iterator_concept` denotes `random_access_iterator_tag`.
 - .2) — Otherwise, if *all-bidirectional*<Const, Views...> is modeled, then `iterator_concept` denotes `bidirectional_iterator_tag`.
 - .3) — Otherwise, if *all-forward*<Const, Views...> is modeled, then `iterator_concept` denotes `forward_iterator_tag`.
 - .4) — Otherwise, `iterator_concept` denotes `input_iterator_tag`.
- 2 `iterator::iterator_category` is present if and only if *all-forward*<Const, Views...> is modeled.
- 3 If the invocation of any non-const member function of *iterator* exits via an exception, the iterator acquires a singular value.

```
constexpr explicit iterator(tuple-or-pair<iterator_t<maybe-const<Const, Views>>...>
    current);
```

- 4 *Effects*: Initializes `current_` with `std::move(current)`.

```
constexpr iterator(iterator<!Const> i)
    requires Const && (convertible_to<iterator_t<Views>, iterator_t<maybe-const<Const,
        Views>>> && ...);
```

- 5 *Effects*: Initializes `current_` with `std::move(i.current_)`.

```
constexpr auto operator*() const;
```

- 6 *Effects*: Equivalent to:

```
return tuple-transform([](auto& i) -> decltype(auto) { return
    *i; }, current_);
```

```
constexpr iterator& operator++();
```

- 7 *Effects*: Equivalent to:

```
tuple-for-each([](auto& i) { ++i; }, current_);
return *this;
```

```
constexpr void operator++(int);
```

- 8 *Effects*: Equivalent to `++*this`.

```
constexpr iterator operator++(int) requires all-forward<Const, Views...>;
```

- 9 *Effects*: Equivalent to:

```
auto tmp = *this;
++*this;
return tmp;
```

```
constexpr iterator& operator--() requires all-bidirectional<Const, Views...>;
```

10 *Effects:* Equivalent to:

```
tuple-for-each([](auto& i) { --i; }, current_);  
return *this;
```

```
constexpr iterator operator--(int) requires all-bidirectional<Const, Views...>;
```

11 *Effects:* Equivalent to:

```
auto tmp = *this;  
--*this;  
return tmp;
```

```
constexpr iterator& operator+=(difference_type x)  
requires all-random-access<Const, Views...>;
```

12 *Effects:* Equivalent to:

```
tuple-for-each([&<class I>(I& i) { i += iter_difference_t<I>  
    (x); }, current_);  
return *this;
```

```
constexpr iterator& operator--(difference_type x)  
requires all-random-access<Const, Views...>;
```

13 *Effects:* Equivalent to:

```
tuple-for-each([&<class I>(I& i) { i -= iter_difference_t<I>  
    (x); }, current_);  
return *this;
```

```
constexpr auto operator[](difference_type n) const  
requires all-random-access<Const, Views...>;
```

14 *Effects:* Equivalent to:

```
return tuple-transform([&<class I>(I& i) -> decltype(auto) {  
    return i[iter_difference_t<I>(n)];  
}, current_);
```

```
friend constexpr bool operator==(const iterator& x, const iterator& y)  
requires (equality_comparable<iterator_t<maybe-const<Const, Views...>> && ...);
```

15 *Returns:*

- (15.1) — `x.current_ == y.current_` if `all-bidirectional<Const, Views...>` is `true`.
- (15.2) — Otherwise, `true` if there exists an integer $0 \leq i < \text{sizeof}...(Views)$ such that `bool(std::get<i>(x.current_) == std::get<i>(y.current_))` is `true`. [Note 1: This allows `zip_view` to model `common_range` when all constituent views model `common_range`. — end note]
- (15.3) — Otherwise, `false`.

```
friend constexpr bool operator<(const iterator& x, const iterator& y)  
requires all-random-access<Const, Views...>;
```

16 Returns: `x.current_ < y.current_`.

```
friend constexpr bool operator<(const iterator& x, const iterator& y)
requires all-random-access<Const, Views...>;
```

17 Effects: Equivalent to: `return y < x;`

```
friend constexpr bool operator<=(const iterator& x, const iterator& y)
requires all-random-access<Const, Views...>;
```

18 Effects: Equivalent to: `return !(y < x);`

```
friend constexpr bool operator>=(const iterator& x, const iterator& y)
requires all-random-access<Const, Views...>;
```

19 Effects: Equivalent to: `return !(x < y);`

```
friend constexpr auto operator<=>(const iterator& x, const iterator& y)
requires all-random-access<Const, Views...> &&
(three_way_comparable<iterator_t<maybe-const<Const, Views>>> && ...);
```

20 Returns: `x.current_ <=> y.current_`.

```
friend constexpr iterator operator+(const iterator& i, difference_type n)
requires all-random-access<Const, Views...>;
friend constexpr iterator operator+(difference_type n, const iterator& i)
requires all-random-access<Const, Views...>;
```

21 Effects: Equivalent to:

```
auto r = i;
r += n;
return r;
```

```
friend constexpr iterator operator-(const iterator& i, difference_type n)
requires all-random-access<Const, Views...>;
```

22 Effects: Equivalent to:

```
auto r = i;
r -= n;
return r;
```

```
friend constexpr difference_type operator-(const iterator& x, const iterator& y)
requires (sized_sentinel_for<iterator_t<maybe-const<Const, Views>>,
iterator_t<maybe-const<Const, Views>>> && ...);
```

23 Let $DIST(i)$ be `difference_type(std::get<i>(x.current_) - std::get<i>(y.current_))`.

24 Returns: The value with the smallest absolute value among $DIST(n)$ for all integers $0 \leq n < \text{sizeof}...(Views)$.

```
friend constexpr auto iter_move(const iterator& i)
noexcept(see below);
```

25 *Effects:* Equivalent to:

```
return tuple-transform(ranges::iter_move, i.current_);
```

26 *Remarks:* The expression within `noexcept` is equivalent to

```
(noexcept(ranges::iter_move(declval<iterator_t<maybe-  
const<Const, Views>> const&>())) && ...) &&  
(is_nothrow_move_constructible_v<range_rvalue_reference_t<maybe-  
const<Const, Views>>> && ...))
```

```
friend constexpr void iter_swap(const iterator& l, const iterator& r)  
noexcept(see below)  
requires (indirectly_swappable<iterator_t<maybe-const<Const, Views>>> && ...);
```

27 *Effects:* For every integer $0 \leq i < \text{sizeof}...(Views)$, performs
`ranges::iter_swap(std::get<i>(l.current_), std::get<i>(r.current_))`.

28 *Remarks:* The expression within `noexcept` is equivalent to the logical AND of the following expressions:

```
noexcept(ranges::iter_swap(std::get<i>(l.current_), std::get<i>  
(r.current_)))
```

for every integer $0 \leq i < \text{sizeof}...(Views)$.

§ 24.7.?.4 Class template `zip_view::sentinel` [range.zip.sentinel]

```
namespace std::ranges {  
    template<input_range... Views>  
        requires (view<Views> && ...) && (sizeof...(Views) > 0)  
    template<bool Const>  
    class zip_view<Views...>::sentinel {  
        tuple-or-pair<sentinel_t<maybe-const<Const, Views>>...> end_;  
        // exposition only  
        constexpr explicit sentinel(tuple-or-pair<sentinel_t<maybe-const<Const,  
            Views>>...> end);    // exposition only  
    public:  
        sentinel() = default;  
        constexpr sentinel(sentinel<!Const> i)  
            requires Const && (convertible_to<sentinel_t<Views>, sentinel_t<maybe-  
                const<Const, Views>>> && ...);  
  
        template<bool OtherConst>  
            requires (sentinel_for<sentinel_t<maybe-const<Const, Views>>, iterator_t<maybe-  
                const<OtherConst, Views>>> && ...)  
        friend constexpr bool operator==(const iterator<OtherConst>& x, const sentinel&  
            y);  
  
        template<bool OtherConst>  
            requires (sized_sentinel_for<sentinel_t<maybe-const<Const, Views>>, iterator_t<maybe-const<OtherConst, Views>>> && ...)  
        friend constexpr common_type_t<range_difference_t<maybe-const<OtherConst, Views>>...>  
            operator-(const iterator<OtherConst>& x, const sentinel& y);
```



```

template<bool OtherConst>
    requires (sized_sentinel_for<sentinel_t<maybe-const<Const, Views>>,
        iterator_t<maybe-const<OtherConst, Views>>> && ...)
    friend constexpr common_type_t<range_difference_t<maybe-const<OtherConst,
        Views>>...>
        operator-(const sentinel& y, const iterator<OtherConst>& x);
};
}

```

```

constexpr explicit sentinel(tuple-or-pair<sentinel_t<maybe-const<Const, Views>>...>
    end);

```

1 *Effects:* Initializes `end_` with `end`.

```

constexpr sentinel(sentinel<!Const> i)
    requires Const && (convertible_to<sentinel_t<Views>, sentinel_t<maybe-const<Const,
        Views>>> && ...);

```

2 *Effects:* Initializes `end_` with `std::move(i.end_)`.

```

template<bool OtherConst>
    requires (sentinel_for<sentinel_t<maybe-const<Const, Views>>, iterator_t<maybe-
        const<OtherConst, Views>>> && ...)
    friend constexpr bool operator==(const iterator<OtherConst>& x, const sentinel& y);

```

3 *Returns:* `true` if there exists an integer $0 \leq i < \text{sizeof}...(Views)$ such that `bool(std::get<i>(x.current_) == std::get<i>(y.end_))` is `true`. Otherwise, `false`.

```

template<bool OtherConst>
    requires (sized_sentinel_for<sentinel_t<maybe-const<Const, Views>>,
        iterator_t<maybe-const<OtherConst, Views>>> && ...)
    friend constexpr common_type_t<range_difference_t<maybe-const<OtherConst, Views>>...>
        operator-(const iterator<OtherConst>& x, const sentinel& y);

```

4 Let D be the return type. Let $DIST(i)$ be $D(\text{std::get}<i>(x.current_) - \text{std::get}<i>(y.end_))$.

5 *Returns:* The value with the smallest absolute value among $DIST(n)$ for all integers $0 \leq n < \text{sizeof}...(Views)$.

```

template<bool OtherConst>
    requires (sized_sentinel_for<sentinel_t<maybe-const<Const, Views>>,
        iterator_t<maybe-const<OtherConst, Views>>> && ...)
    friend constexpr common_type_t<range_difference_t<maybe-const<OtherConst, Views>>...>
        operator-(const sentinel& y, const iterator<OtherConst>& x);

```

6 *Effects:* Equivalent to `return -(x - y);`

§ 5.7 zip_transform

Add the following subclause to 24.7 [\[range.adaptors\]](#).

§ 24.7.? Zip transform view [\[range.zip.transform\]](#)

§ 24.7.?.1 Overview [range.zip.transform.overview]

- 1 zip_transform_view takes an invocable object and any number of views and produces a view whose M^{th} element is the result of applying the invocable object to the M^{th} elements of all views.
- 2 The name views::zip_transform denotes a customization point object (16.3.3.3.6 [customization.point.object]). Let F be a subexpression, and let Es... be a pack of subexpressions.
 - .1) — If Es is an empty pack, let FD be decay_t<decltype((F))>.
 - .1) — If copy_constructible<FD> && regular_invocable<FD&> is false, or if decay_t<invoke_result_t<FD&>> is not an object type, views::zip_transform(F, Es...) is ill-formed.
 - .2) — Otherwise, the expression views::zip_transform(F, Es...) is expression-equivalent to ((void)F, decay-copy(views::empty<decay_t<invoke_result_t<FD&>>>)).
 - .2) — Otherwise, the expression views::zip_transform(F, Es...) is expression-equivalent to zip_transform_view(F, Es...).

[Example 1:

```
vector v1 = {1, 2};
vector v2 = {4, 5, 6};

for (auto i : views::zip_transform(plus(), v1, v2)) {
    cout << i << ' '; // prints: 5 7
}
```

— end example]

§ 24.7.?.2 Class template zip_transform_view [range.zip.transform.view]

```
namespace std::ranges {

template<copy_constructible F, input_range... Views>
    requires (view<Views> && ...) && (sizeof...(Views) > 0) && is_object_v<F> &&
        regular_invocable<F&, range_reference_t<Views>...> &&
        can_reference<invoke_result_t<F&, range_reference_t<Views>...>>
class zip_transform_view : public view_interface<zip_transform_view<F, Views...>> {
    copyable_box<F> fun_; // exposition only
    zip_view<Views...> zip_; // exposition only

    using InnerView = zip_view<Views...>; // exposition only
    template<bool Const>
    using zipperator = iterator_t<maybe_const<Const, InnerView>>; // exposition only
    template<bool Const>
    using zentinel = sentinel_t<maybe_const<Const, InnerView>>; // exposition only

    template<bool> class iterator; // exposition only
    template<bool> class sentinel; // exposition only

public:
    zip_transform_view() = default;
```

```

constexpr explicit zip_transform_view(F fun, Views... views);

constexpr auto begin() {
    return iterator<false>(*this, zip_.begin());
}

constexpr auto begin() const
    requires range<const InnerView> &&
        regular_invocable<const F&, range_reference_t<const Views>...>
{
    return iterator<true>(*this, zip_.begin());
}

constexpr auto end() {
    if constexpr (common_range<InnerView>) {
        return iterator<false>(*this, zip_.end());
    }
    else {
        return sentinel<false>(zip_.end());
    }
}

constexpr auto end() const
    requires range<const InnerView> &&
        regular_invocable<const F&, range_reference_t<const Views>...>
{
    if constexpr (common_range<const InnerView>) {
        return iterator<true>(*this, zip_.end());
    }
    else {
        return sentinel<true>(zip_.end());
    }
}

constexpr auto size() requires sized_range<InnerView> {
    return zip_.size();
}

constexpr auto size() const requires sized_range<const InnerView> {
    return zip_.size();
}
};

template<class F, class... Rs>
zip_transform_view(F, Rs&&...) -> zip_transform_view<F, views::all_t<Rs>...>;
}

constexpr explicit zip_transform_view(F fun, Views... views);

```

¹ *Effects:* Initializes *fun_* with `std::move(fun)` and *zip_* with `std::move(views)....`

§ 24.7.3.3 Class template `zip_transform_view::iterator` [range.zip.transform.iterator]

```

namespace std::ranges {
    template<copy_constructible F, input_range... Views>
        requires (view<Views> && ...) && (sizeof...(Views) > 0) && is_object_v<F> &&
            regular_invocable<F&, range_reference_t<Views>...> &&
            can-reference<invoke_result_t<F&, range_reference_t<Views>...>>
    template<bool Const>
    class zip_transform_view<F, Views...>::iterator {
        using Parent = maybe-const<Const, zip_transform_view>;           // exposition only
        using Base = maybe-const<Const, InnerView>;                     // exposition only
        Parent* parent_ = nullptr;                                       // exposition only
        ziperator<Const> inner_;                                           // exposition only

        constexpr iterator(Parent& parent, ziperator<Const> inner); // exposition only

    public:
        using iterator_category = see below;                             // not always present
        using iterator_concept = typename ziperator<Const>::iterator_concept;
        using value_type =
            remove_cvref_t<invoke_result_t<maybe-const<Const, F>&, range_reference_t<maybe-
                const<Const, Views>>...>>;
        using difference_type = range_difference_t<Base>;

        iterator() = default;
        constexpr iterator(iterator<!Const> i)
            requires Const && convertible_to<ziperator<false>, ziperator<Const>>;

        constexpr decltype(auto) operator*() const noexcept(see below);
        constexpr iterator& operator++();
        constexpr void operator++(int);
        constexpr iterator operator++(int) requires forward_range<Base>;

        constexpr iterator& operator--() requires bidirectional_range<Base>;
        constexpr iterator operator--(int) requires bidirectional_range<Base>;

        constexpr iterator& operator+=(difference_type x) requires
            random_access_range<Base>;
        constexpr iterator& operator-=(difference_type x) requires
            random_access_range<Base>;

        constexpr decltype(auto) operator[](difference_type n) const requires
            random_access_range<Base>;

        friend constexpr bool operator==(const iterator& x, const iterator& y)
            requires equality_comparable<ziperator<Const>>;

        friend constexpr bool operator<(const iterator& x, const iterator& y)
            requires random_access_range<Base>;
        friend constexpr bool operator>(const iterator& x, const iterator& y)
            requires random_access_range<Base>;
        friend constexpr bool operator<=(const iterator& x, const iterator& y)
            requires random_access_range<Base>;
        friend constexpr bool operator>=(const iterator& x, const iterator& y)
            requires random_access_range<Base>;
        friend constexpr auto operator<=>(const iterator& x, const iterator& y)
            requires random_access_range<Base> && three_way_comparable<ziperator<Const>>;

```

```

friend constexpr iterator operator+(const iterator& i, difference_type n)
    requires random_access_range<Base>;
friend constexpr iterator operator+(difference_type n, const iterator& i)
    requires random_access_range<Base>;
friend constexpr iterator operator-(const iterator& i, difference_type n)
    requires random_access_range<Base>;
friend constexpr difference_type operator-(const iterator& x, const iterator& y)
    requires sized_sentinel_for<ziperator<Const>, ziperator<Const>>;
};
}

```

1 The member *typedef-name* `iterator::iterator_category` is defined if and only if *Base* models `forward_range`. In that case, `iterator::iterator_category` is defined as follows:

- .1) — If `invoke_result_t<maybe-const<Const, F>&, range_reference_t<maybe-const<Const, Views>>...>` is not an lvalue reference, `iterator_category` denotes `input_iterator_tag`.
- .2) — Otherwise, let `Cs...` denote the pack of types `iterator_traits<iterator_t<maybe-const<Const, Views>>>::iterator_category...`
 - .1) — If `(derived_from<Cs, random_access_iterator_tag> && ...)` is `true`, `iterator_category` denotes `random_access_iterator_tag`.
 - .2) — Otherwise, if `(derived_from<Cs, bidirectional_iterator_tag> && ...)` is `true`, `iterator_category` denotes `bidirectional_iterator_tag`.
 - .3) — Otherwise, if `(derived_from<Cs, forward_iterator_tag> && ...)` is `true`, `iterator_category` denotes `forward_iterator_tag`.
 - .4) — Otherwise, `iterator_category` denotes `input_iterator_tag`.

```
constexpr iterator(Parent& parent, ziperator<Const> inner);
```

2 *Effects*: Initializes `parent_` with `addressof(parent)` and `inner_` with `std::move(inner)`.

```
constexpr iterator(iterator<!Const> i)
    requires Const && convertible_to<ziperator<false>, ziperator<Const>>;
```

3 *Effects*: Initializes `parent_` with `i.parent_` and `inner_` with `std::move(i.inner_)`.

```
constexpr decltype(auto) operator*() const noexcept(see below);
```

4 *Effects*: Equivalent to:

```

return apply([&](const auto&... iters) -> decltype(auto) {
    return invoke(*parent_->fun_, *iters...);
}, inner_.current_);

```

5 *Remarks*: Let `Is` be the pack `0, 1, ..., (sizeof...(Views)-1)`. The expression within `noexcept` is equivalent to `noexcept(invoke(*parent_->fun_, *std::get<Is>(inner_.current_)...))`.

```
constexpr iterator& operator++();
```

6 *Effects*: Equivalent to:

```
++inner_;  
return *this;
```

```
constexpr void operator++(int);
```

7 *Effects:* Equivalent to `++*this`.

```
constexpr iterator operator++(int) requires forward_range<Base>;
```

8 *Effects:* Equivalent to:

```
auto tmp = *this;  
++*this;  
return tmp;
```

```
constexpr iterator& operator--() requires bidirectional_range<Base>;
```

9 *Effects:* Equivalent to:

```
--inner_;  
return *this;
```

```
constexpr iterator operator--(int) requires bidirectional_range<Base>;
```

10 *Effects:* Equivalent to:

```
auto tmp = *this;  
--*this;  
return tmp;
```

```
constexpr iterator& operator+=(difference_type x)  
requires random_access_range<Base>;
```

11 *Effects:* Equivalent to:

```
inner_ += x;  
return *this;
```

```
constexpr iterator& operator-=(difference_type x)  
requires random_access_range<Base>;
```

12 *Effects:* Equivalent to:

```
inner_ -= x;  
return *this;
```

```
constexpr decltype(auto) operator[](difference_type n) const  
requires random_access_range<Base>;
```

13 *Effects:* Equivalent to:

```
return apply([&<class...Is>(const Is&... iters) ->  
    decltype(auto) {  
        return invoke(*parent_->fun_, iters[iter_difference_t<Is>  
            (n)]...);  
    }, inner_.current_);
```

```

friend constexpr bool operator==(const iterator& x, const iterator& y)
    requires equality_comparable<ziperator<Const>>;
friend constexpr bool operator<(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator>(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator<=(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator>=(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr auto operator<=>(const iterator& x, const iterator& y)
    requires random_access_range<Base> && three_way_comparable<ziperator<Const>>;

```

14 Let *op* be the operator.

15 *Effects*: Equivalent to: `return x.inner_ op y.inner_;`

```

friend constexpr iterator operator+(const iterator& i, difference_type n)
    requires random_access_range<Base>;
friend constexpr iterator operator+(difference_type n, const iterator& i)
    requires random_access_range<Base>;

```

16 *Effects*: Equivalent to: `return iterator(*i.parent_, i.inner_ + n);`

```

friend constexpr iterator operator-(const iterator& i, difference_type n)
    requires random_access_range<Base>;

```

17 *Effects*: Equivalent to: `return iterator(*i.parent_, i.inner_ - n);`

```

friend constexpr difference_type operator-(const iterator& x, const iterator& y)
    requires sized_sentinel_for<ziperator<Const>, ziperator<Const>>;

```

18 *Effects*: Equivalent to: `return x.inner_ - y.inner_;`

§ 24.7.?.4 Class template `zip_transform_view::sentinel` [range.zip.transform.sentinel]

```

namespace std::ranges {
    template<copy_constructible F, input_range... Views>
        requires (view<Views> && ...) && (sizeof...(Views) > 0) && is_object_v<F> &&
            regular_invocable<F&, range_reference_t<Views>...> &&
            can_reference<invoke_result_t<F&, range_reference_t<Views>...>>
        template<bool Const>
        class zip_transform_view<F, Views...>::sentinel {
            zentinel<Const> inner_; // exposition only
            constexpr explicit sentinel(zentinel<Const> inner); // exposition only
        public:
            sentinel() = default;
            constexpr sentinel(sentinel<!Const> i)
                requires Const && convertible_to<zentinel<false>, zentinel<Const>>;

            template<bool OtherConst>
                requires sentinel_for<zentinel<Const>, ziperator<OtherConst>>
            friend constexpr bool operator==(const iterator<OtherConst>& x, const sentinel&
                y);

```

```

template<bool OtherConst>
    requires sized_sentinel_for<z Sentinel<Const>, ziperator<OtherConst>>
friend constexpr range_difference_t<maybe-const<OtherConst, InnerView>>
    operator-(const iterator<OtherConst>& x, const Sentinel& y);

template<bool OtherConst>
    requires sized_sentinel_for<z Sentinel<Const>, ziperator<OtherConst>>
friend constexpr range_difference_t<maybe-const<OtherConst, InnerView>>
    operator-(const Sentinel& x, const iterator<OtherConst>& y);
};
}

```

```
constexpr explicit Sentinel(z Sentinel<Const> inner);
```

¹ *Effects:* Initializes `inner_` with `inner`.

```
constexpr Sentinel(Sentinel<!Const> i)
    requires Const && convertible_to<z Sentinel<false>, Sentinel<Const>>;
```

² *Effects:* Initializes `inner_` with `std::move(i.inner_)`.

```

template<bool OtherConst>
    requires Sentinel_for<z Sentinel<Const>, ziperator<OtherConst>>
friend constexpr bool operator==(const iterator<OtherConst>& x, const Sentinel& y);

```

³ *Effects:* Equivalent to `return x.inner_ == y.inner_;`

```

template<bool OtherConst>
    requires sized_sentinel_for<z Sentinel<Const>, ziperator<OtherConst>>
friend constexpr range_difference_t<maybe-const<OtherConst, InnerView>>
    operator-(const iterator<OtherConst>& x, const Sentinel& y);

template<bool OtherConst>
    requires sized_sentinel_for<z Sentinel<Const>, ziperator<OtherConst>>
friend constexpr range_difference_t<maybe-const<OtherConst, InnerView>>
    operator-(const Sentinel& x, const iterator<OtherConst>& y);

```

⁴ *Effects:* Equivalent to `return x.inner_ - y.inner_;`

§ 5.8 adjacent

Add the following subclause to 24.7 [\[range.adaptors\]](#).

§ 24.7.? Adjacent view [\[range.adjacent\]](#)

§ 24.7.?.1 Overview [\[range.adjacent.overview\]](#)

¹ `adjacent_view` takes a view and produces a view whose M^{th} element is a tuple of references to the M^{th} through $(M + N - 1)^{\text{th}}$ elements of the original view. If the original view has fewer than N elements, the resulting view is empty.

2 The name `views::adjacent<N>` denotes a range adaptor object (24.7.2 [\[range.adaptor.object\]](#)). Given a subexpression `E` and a constant expression `N`, the expression `views::adjacent<N>(E)` is expression-equivalent to

- .1) — `((void)E, decay-copy(views::empty<tuple<>>))` if `N` is equal to `0`,
- .2) — otherwise, `adjacent_view<views::all_t<decltype((E))>, N>(E)`.

[Example 1:

```
vector v = {1, 2, 3, 4};

for (auto i : v | views::adjacent<2>) {
    cout << '(' << i.first << ', ' << i.second << ") "; // prints: (1, 2) (2, 3) (3, 4)
}
```

— end example]

3 Define `REPEAT(T, N)` as a pack of `N` types, each of which denotes the same type as `T`.

§ 24.7.?.2 Class template `adjacent_view` [\[range.adjacent.view\]](#)

```
namespace std::ranges {

template<forward_range V, size_t N>
    requires view<V> && (N > 0)
class adjacent_view : public view_interface<adjacent_view<V, N>>{
    V base_ = V(); // exposition only

    template<bool> class iterator; // exposition only
    template<bool> class sentinel; // exposition only

    struct as-sentinel{}; // exposition only

public:
    adjacent_view() requires default_initializable<V> = default;
    constexpr explicit adjacent_view(V base);

    constexpr auto begin() requires (!simple-view<V>) {
        return iterator<false>(ranges::begin(base_), ranges::end(base_));
    }

    constexpr auto begin() const requires range<const V> {
        return iterator<true>(ranges::begin(base_), ranges::end(base_));
    }

    constexpr auto end() requires (!simple-view<V>) {
        if constexpr (common_range<V>) {
            return iterator<false>(as-sentinel{}, ranges::begin(base_), ranges::end(base_));
        }
        else {
            return sentinel<false>(ranges::end(base_));
        }
    }
}
```



```
constexpr auto end() requires range<const V> {
    if constexpr (common_range<const V>) {
        return iterator<true>(as-sentinel{}, ranges::begin(base_), ranges::end(base_));
    }
    else {
        return sentinel<true>(ranges::end(base_));
    }
}

constexpr auto size() requires sized_range<V>;
constexpr auto size() const requires sized_range<const V>;
};
}
```

```
constexpr explicit adjacent_view(V base);
```

- 1 *Effects:* Initializes `base_` with `std::move(base)`.

```
constexpr auto size() requires sized_range<V>;
constexpr auto size() const requires sized_range<const V>;
```

- 2 *Effects:* Equivalent to:

```
using ST = decltype(ranges::size(base_));
using CT = common_type_t<ST, size_t>;
auto sz = static_cast<CT>(ranges::size(base_));
sz -= std::min<CT>(sz, N - 1);
return static_cast<ST>(sz);
```

§ 24.7.3 Class template `adjacent_view::iterator` [range.adjacent.iterator]

```
namespace std::ranges {
    template<forward_range V, size_t N>
        requires view<V> && (N > 0)
    template<bool Const>
    class adjacent_view<V, N>::iterator {
        using Base = maybe_const<Const, V>; //
        exposition only
        array<iterator_t<Base>, N> current_ = array<iterator_t<Base>, N>(); //
        exposition only
        constexpr iterator(iterator_t<Base> first, sentinel_t<Base> last); //
        exposition only
        constexpr iterator(as-sentinel, iterator_t<Base> first, iterator_t<Base> last); //
        exposition only
    public:
        using iterator_category = input_iterator_tag;
        using iterator_concept = see below;
        using value_type = tuple-or-pair<REPEAT(range_value_t<Base>, N)...>;
        using difference_type = range_difference_t<Base>;

        iterator() = default;
        constexpr iterator(iterator<!Const> i)
            requires Const && convertible_to<iterator_t<V>, iterator_t<Base>>;

        constexpr auto operator*() const;
```

```

constexpr iterator& operator++();
constexpr iterator operator++(int);

constexpr iterator& operator--() requires bidirectional_range<Base>;
constexpr iterator operator--(int) requires bidirectional_range<Base>;

constexpr iterator& operator+=(difference_type x)
    requires random_access_range<Base>;
constexpr iterator& operator-=(difference_type x)
    requires random_access_range<Base>;

constexpr auto operator[](difference_type n) const
    requires random_access_range<Base>;

friend constexpr bool operator==(const iterator& x, const iterator& y);

friend constexpr bool operator<(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator>(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator<=(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator>=(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr auto operator<=>(const iterator& x, const iterator& y)
    requires random_access_range<Base> &&
        three_way_comparable<iterator_t<Base>>;

friend constexpr iterator operator+(const iterator& i, difference_type n)
    requires random_access_range<Base>;
friend constexpr iterator operator+(difference_type n, const iterator& i)
    requires random_access_range<Base>;
friend constexpr iterator operator-(const iterator& i, difference_type n)
    requires random_access_range<Base>;
friend constexpr difference_type operator-(const iterator& x, const iterator& y)
    requires sized_sentinel_for<iterator_t<Base>, iterator_t<Base>>;

friend constexpr auto iter_move(const iterator& i) noexcept(see below);

friend constexpr void iter_swap(const iterator& l, const iterator& r)
    noexcept(see below)
    requires indirectly_swappable<iterator_t<Base>>;
};
}

```

1 `iterator::iterator_concept` is defined as follows:

- .1) — If *Base* models `random_access_range`, then `iterator_concept` denotes `random_access_iterator_tag`.
- .2) — Otherwise, if *Base* models `bidirectional_range`, then `iterator_concept` denotes `bidirectional_iterator_tag`.
- .3) — Otherwise, `iterator_concept` denotes `forward_iterator_tag`.

- 2 If the invocation of any non-const member function of *iterator* exits via an exception, the iterator acquires a singular value.

```
constexpr iterator(iterator_t<Base> first, sentinel_t<Base> last);
```

- 3 *Postconditions:* `current_[0] == first` is `true`, and for every integer $1 \leq i < N$, `current_[i] == ranges::next(current_[i-1], 1, last)` is `true`.

```
constexpr iterator(as-sentinel, iterator_t<Base> first, iterator_t<Base> last);
```

- 4 *Postconditions:* If *Base* does not model `bidirectional_range`, each element of `current_` is equal to `last`. Otherwise, `current_[N-1] == last` is `true`, and for every integer $0 \leq i < (N - 1)$, `current_[i] == ranges::prev(current_[i+1], 1, first)` is `true`.

```
constexpr iterator(iterator<!Const> i)  
requires Const && (convertible_to<iterator_t<V>, iterator_t<Base>>);
```

- 5 *Effects:* Initializes each element of `current_` with the corresponding element of `i.current_` as an xvalue.

```
constexpr auto operator*() const;
```

- 6 *Effects:* Equivalent to:

```
return tuple-transform([](auto& i) -> decltype(auto) { return  
    *i; }, current_);
```

```
constexpr iterator& operator++();
```

- 7 *Preconditions:* `current_.back()` is incrementable.

- 8 *Postconditions:* Each element of `current_` is equal to `ranges::next(i)`, where `i` is the value of that element before the call.

- 9 *Returns:* `*this`.

```
constexpr iterator operator++(int);
```

- 10 *Effects:* Equivalent to:

```
auto tmp = *this;  
++*this;  
return tmp;
```

```
constexpr iterator& operator--() requires bidirectional_range<Base>;
```

- 11 *Preconditions:* `current_.front()` is decrementable.

- 12 *Postconditions:* Each element of `current_` is equal to `ranges::prev(i)`, where `i` is the value of that element before the call.

- 13 *Returns:* `*this`.

```
constexpr iterator operator--(int) requires bidirectional_range<Base>;
```

14 *Effects:* Equivalent to:

```
auto tmp = *this;  
--*this;  
return tmp;
```

```
constexpr iterator& operator+=(difference_type x)  
requires random_access_range<Base>;
```

15 *Preconditions:* `current_.back() + x` has well-defined behavior.

16 *Postconditions:* Each element of `current_` is equal to $i + x$, where i is the value of that element before the call.

17 *Returns:* `*this`.

```
constexpr iterator& operator-=(difference_type x)  
requires random_access_range<Base>;
```

18 *Preconditions:* `current_.front() - x` has well-defined behavior.

19 *Postconditions:* Each element of `current_` is equal to $i - x$, where i is the value of that element before the call.

20 *Returns:* `*this`.

```
constexpr auto operator[](difference_type n) const  
requires random_access_range<Base>;
```

21 *Effects:* Equivalent to:

```
return tuple-transform([&](auto& i) -> decltype(auto) { return  
    i[n]; }, current_);
```

```
friend constexpr bool operator==(const iterator& x, const iterator& y);
```

22 *Returns:* `x.current_.back() == y.current_.back()`.

```
friend constexpr bool operator<(const iterator& x, const iterator& y)  
requires random_access_range<Base>;
```

23 *Returns:* `x.current_.back() < y.current_.back()`.

```
friend constexpr bool operator>(const iterator& x, const iterator& y)  
requires random_access_range<Base>;
```

24 *Effects:* Equivalent to: `return y < x;`

```
friend constexpr bool operator<=(const iterator& x, const iterator& y)  
requires random_access_range<Base>;
```

25 *Effects:* Equivalent to: `return !(y < x);`

```
friend constexpr bool operator>=(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
```

26 *Effects:* Equivalent to: `return !(x < y);`

```
friend constexpr auto operator<=>(const iterator& x, const iterator& y)
    requires random_access_range<Base> &&
        three_way_comparable<iterator_t<Base>>;
```

27 *Returns:* `x.current_.back() <=> y.current_.back()`.

```
friend constexpr iterator operator+(const iterator& i, difference_type n)
    requires random_access_range<Base>;
friend constexpr iterator operator+(difference_type n, const iterator& i)
    requires random_access_range<Base>;
```

28 *Effects:* Equivalent to:

```
auto r = i;
r += n;
return r;
```

```
friend constexpr iterator operator-(const iterator& i, difference_type n)
    requires random_access_range<Base>;
```

29 *Effects:* Equivalent to:

```
auto r = i;
r -= n;
return r;
```

```
friend constexpr difference_type operator-(const iterator& x, const iterator& y)
    requires sized_sentinel_for<iterator_t<Base>, iterator_t<Base>>;
```

30 *Effects:* Equivalent to: `return x.current_.back() - y.current_.back();`

```
friend constexpr auto iter_move(const iterator& i) noexcept(see below);
```

31 *Effects:* Equivalent to:

```
return tuple-transform(ranges::iter_move, i.current_);
```

32 *Remarks:* The expression within `noexcept` is equivalent to

```
noexcept(ranges::iter_move(declval<iterator_t<Base> const&>()))
    &&
    is_nothrow_move_constructible_v<range_rvalue_reference_t<Base>>
```

```
friend constexpr void iter_swap(const iterator& l, const iterator& r)
    noexcept(see below)
    requires indirectly_swappable<iterator_t<Base>>;
```

33 *Preconditions:* None of the iterators in `l.current_` is equal to an iterator in `r.current_`.

34 *Effects:* For every integer $0 \leq i < N$, performs `ranges::iter_swap(l.current_[i], r.current_[i])`.

35 *Remarks:* The expression within `noexcept` is equivalent to:

```
noexcept(ranges::iter_swap(declval<iterator_t<Base>>(),
                          declval<iterator_t<Base>>()))
```

§ 24.7.?.4 Class template `adjacent_view::sentinel` [range.adjacent.sentinel]

```
namespace std::ranges {
    template<forward_range V, size_t N>
        requires view<V> && (N > 0)
    template<bool Const>
    class adjacent_view<V, N>::sentinel {
        using Base = maybe_const<Const, V>; // exposition only
        sentinel_t<Base> end_ = sentinel_t<Base>(); // exposition only
        constexpr explicit sentinel(sentinel_t<Base> end); // exposition only
    public:
        sentinel() = default;
        constexpr sentinel(sentinel<!Const> i)
            requires Const && convertible_to<sentinel_t<V>, sentinel_t<Base>>;

        template<bool OtherConst>
            requires sentinel_for<sentinel_t<Base>, iterator_t<maybe_const<OtherConst, V>>>
        friend constexpr bool operator==(const iterator<OtherConst>& x, const sentinel&
            y);

        template<bool OtherConst>
            requires sized_sentinel_for<sentinel_t<Base>, iterator_t<maybe_const<OtherConst, V>>>
        friend constexpr range_difference_t<maybe_const<OtherConst, V>>
            operator-(const iterator<OtherConst>& x, const sentinel& y);

        template<bool OtherConst>
            requires sized_sentinel_for<sentinel_t<Base>, iterator_t<maybe_const<OtherConst, V>>>
        friend constexpr range_difference_t<maybe_const<OtherConst, V>>
            operator-(const sentinel& y, const iterator<OtherConst>& x);
    };
}
```

```
constexpr explicit sentinel(sentinel_t<Base> end);
```

1 *Effects:* Initializes `end_` with `end`.

```
constexpr sentinel(sentinel<!Const> i)
    requires Const && convertible_to<sentinel_t<V>, sentinel_t<Base>>;
```

2 *Effects:* Initializes `end_` with `std::move(i.end_)`.

```
template<bool OtherConst>
    requires sentinel_for<sentinel_t<Base>, iterator_t<maybe_const<OtherConst, V>>>
friend constexpr bool operator==(const iterator<OtherConst>& x, const sentinel& y);
```

3 *Effects:* Equivalent to: `return x.current_.back() == y.end_;`

```
template<bool OtherConst>
requires sized_sentinel_for<sentinel_t<Base>, iterator_t<maybe-const<OtherConst,
V>>>
friend constexpr range_difference_t<maybe-const<OtherConst, V>>
operator-(const iterator<OtherConst>& x, const sentinel& y);
```

4 *Effects:* Equivalent to: `return x.current_.back() - y.end_;`

```
template<bool OtherConst>
requires sized_sentinel_for<sentinel_t<Base>, iterator_t<maybe-const<OtherConst,
V>>>
friend constexpr range_difference_t<maybe-const<OtherConst, V>>
operator-(const sentinel& y, const iterator<OtherConst>& x);
```

5 *Effects:* Equivalent to: `return y.end_ - x.current_.back();`

§ 5.9 adjacent_transform

Add the following subclause to 24.7 [\[range.adaptors\]](#).

§ 24.7.? Adjacent transform view [\[range.adjacent.transform\]](#)

§ 24.7.?.1 Overview [\[range.adjacent.transform.overview\]](#)

- 1 adjacent_transform_view takes an invocable object and a view and produces a view whose M^{th} element is the result of applying the invocable object to the M^{th} through $(M + N - 1)^{\text{th}}$ elements of the original view. If the original view has fewer than N elements, the resulting view is empty.
- 2 The name `views::adjacent_transform<N>` denotes a range adaptor object (24.7.2 [\[range.adaptor.object\]](#)). Given subexpressions E and F and a constant expression N :
 - .1) — If N is equal to 0, `views::adjacent_transform<N>(E, F)` is expression-equivalent to `((void)E, views::zip_transform(F))`, except that the evaluations of E and F are indeterminately sequenced.
 - .2) — Otherwise, the expression `views::adjacent_transform<N>(E, F)` is expression-equivalent to `adjacent_transform_view<views::all_t<decltype((E))>, decay_t<decltype((F))>, N>(E, F)`.

[Example 1:

```
vector v = {1, 2, 3, 4};

for (auto i : v | views::adjacent_transform<2>(std::multiplies())) {
    cout << i << ' '; // prints: 2 6 12
}
```

— end example]

§ 24.7.?.2 Class template adjacent_transform_view [\[range.adjacent.transform.view\]](#)

```

namespace std::ranges {
    template<forward_range V, copy_constructible F, size_t N>
        requires view<V> && (N > 0) && is_object_v<F> &&
            regular_invocable<F&, REPEAT(range_reference_t<V>, N)...> &&
            can-reference<invoke_result_t<F&, REPEAT(range_reference_t<V>, N)...>>
    class adjacent_transform_view : public view_interface<adjacent_transform_view<V, F,
        N>> {
        copyable_box<F> fun_; // exposition only
        adjacent_view<V, N> inner_; // exposition only

        using InnerView = adjacent_view<V, N>; // exposition only
        template<bool Const>
        using inner_iterator = iterator_t<maybe_const<Const, InnerView>>; // exposition
            only
        template<bool Const>
        using inner_sentinel = sentinel_t<maybe_const<Const, InnerView>>; // exposition
            only

        template<bool> class iterator; // exposition only
        template<bool> class sentinel; // exposition only

    public:
        adjacent_transform_view() = default;

        constexpr explicit adjacent_transform_view(V base, F fun);

        constexpr auto begin() {
            return iterator<false>(*this, inner_.begin());
        }

        constexpr auto begin() const
            requires range<const InnerView> &&
                regular_invocable<const F&, REPEAT(range_reference_t<const V>, N)...>
        {
            return iterator<true>(*this, inner_.begin());
        }

        constexpr auto end() {
            if constexpr (common_range<InnerView>) {
                return iterator<false>(*this, inner_.end());
            }
            else {
                return sentinel<false>(inner_.end());
            }
        }

        constexpr auto end() const
            requires range<const InnerView> &&
                regular_invocable<const F&, REPEAT(range_reference_t<const V>, N)...>
        {
            if constexpr (common_range<const InnerView>) {
                return iterator<true>(*this, inner_.end());
            }
            else {
                return sentinel<true>(inner_.end());
            }
        }
    };
}

```



```

}

constexpr auto size() requires sized_range<InnerView> {
    return inner_.size();
}

constexpr auto size() const requires sized_range<const InnerView> {
    return inner_.size();
}
};
}

```

```
constexpr explicit adjacent_transform_view(V base, F fun);
```

¹ *Effects:* Initializes *fun_* with `std::move(fun)` and *inner_* with `std::move(base)`.

§ 24.7.3.3 Class template `adjacent_transform_view::iterator` [range.adjacent.transform.iterator]

```

namespace std::ranges {
    template<forward_range V, copy_constructible F, size_t N>
        requires view<V> && (N > 0) && is_object_v<F> &&
            regular_invocable<F&, REPEAT(range_reference_t<V>, N)...> &&
            can_reference<invoke_result_t<F&, REPEAT(range_reference_t<V>, N)...>>
    template<bool Const>
    class adjacent_transform_view<F, V...>::iterator {
        using Parent = maybe_const<Const, adjacent_transform_view>;           // exposition
        only
        using Base = maybe_const<Const, V>;                                   // exposition
        only
        Parent* parent_ = nullptr;                                           // exposition
        only
        inner_iterator<Const> inner_;                                         // exposition
        only

        constexpr iterator(Parent& parent, inner_iterator<Const> inner); // exposition
        only

    public:
        using iterator_category = see below;
        using iterator_concept = typename inner_iterator<Const>::iterator_concept;
        using value_type =
            remove_cvref_t<invoke_result_t<maybe_const<Const, F>&,
                REPEAT(range_reference_t<Base>, N)...>>;
        using difference_type = range_difference_t<Base>;

        iterator() = default;
        constexpr iterator(iterator<!Const> i)
            requires Const && convertible_to<inner_iterator<false>, inner_iterator<Const>>;

        constexpr decltype(auto) operator*() const noexcept(see below);
        constexpr iterator& operator++();
        constexpr iterator operator++(int);

        constexpr iterator& operator--() requires bidirectional_range<Base>;
        constexpr iterator operator--(int) requires bidirectional_range<Base>;

```

```

constexpr iterator& operator+=(difference_type x) requires
    random_access_range<Base>;
constexpr iterator& operator-=(difference_type x) requires
    random_access_range<Base>;

constexpr decltype(auto) operator[](difference_type n) const requires
    random_access_range<Base>;

friend constexpr bool operator==(const iterator& x, const iterator& y);

friend constexpr bool operator<(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator>(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator<=(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator>=(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr auto operator<=>(const iterator& x, const iterator& y)
    requires random_access_range<Base> && three_way_comparable<inner-
        iterator<Const>>;

friend constexpr iterator operator+(const iterator& i, difference_type n)
    requires random_access_range<Base>;
friend constexpr iterator operator+(difference_type n, const iterator& i)
    requires random_access_range<Base>;
friend constexpr iterator operator-(const iterator& i, difference_type n)
    requires random_access_range<Base>;
friend constexpr difference_type operator-(const iterator& x, const iterator& y)
    requires sized_sentinel_for<inner-iterator<Const>, inner-iterator<Const>>;
};
}

```

1 The member *typedef-name* `iterator::iterator_category` is defined as follows:

- .1) — If `invoke_result_t<maybe-const<Const, F>&, REPEAT(range_reference_t<Base>, N)...>` is not an lvalue reference, `iterator_category` denotes `input_iterator_tag`.
- .2) — Otherwise, let `C` denote the type `iterator_traits<iterator_t<Base>>::iterator_category`.
 - .1) — If `derived_from<C, random_access_iterator_tag>` is `true`, `iterator_category` denotes `random_access_iterator_tag`.
 - .2) — Otherwise, if `derived_from<C, bidirectional_iterator_tag>` is `true`, `iterator_category` denotes `bidirectional_iterator_tag`.
 - .3) — Otherwise, if `derived_from<C, forward_iterator_tag>` is `true`, `iterator_category` denotes `forward_iterator_tag`.
 - .4) — Otherwise, `iterator_category` denotes `input_iterator_tag`.

```
constexpr iterator(Parent& parent, inner-iterator<Const> inner);
```

- 2 *Effects*: Initializes `parent_` with `addressof(parent)` and `inner_` with `std::move(inner)`.

```
constexpr iterator(iterator<!Const> i)
    requires Const && convertible_to<inner-iterator<false>, inner-iterator<Const>>;
```

3 *Effects:* Initializes *parent_* with *i.parent_* and *inner_* with `std::move(i.inner_)`.

```
constexpr decltype(auto) operator*() const noexcept(see below);
```

4 *Effects:* Equivalent to:

```
return apply([&](const auto&... iters) -> decltype(auto) {
    return invoke(*parent_->fun_, *iters...);
}, inner_.current_);
```

5 *Remarks:* Let *Is* be the pack `0, 1, ..., (N-1)`. The expression within `noexcept` is equivalent to `noexcept(invoke(*parent_->fun_, *std::get<Is>(inner_.current_)...))`.

```
constexpr iterator& operator++();
```

6 *Effects:* Equivalent to:

```
++inner_;
return *this;
```

```
constexpr iterator operator++(int);
```

7 *Effects:* Equivalent to:

```
auto tmp = *this;
++*this;
return tmp;
```

```
constexpr iterator& operator--() requires bidirectional_range<Base>;
```

8 *Effects:* Equivalent to:

```
--inner_;
return *this;
```

```
constexpr iterator operator--(int) requires bidirectional_range<Base>;
```

9 *Effects:* Equivalent to:

```
auto tmp = *this;
--*this;
return tmp;
```

```
constexpr iterator& operator+=(difference_type x)
    requires random_access_range<Base>;
```

10 *Effects:* Equivalent to:

```
inner_ += x;
return *this;
```

```
constexpr iterator& operator--(difference_type x)
    requires random_access_range<Base>;
```

11 *Effects:* Equivalent to:

```
    inner_ -= x;
    return *this;
```

```
constexpr decltype(auto) operator[](difference_type n) const
    requires random_access_range<Base>;
```

12 *Effects:* Equivalent to:

```
    return apply([&](const auto&... iters) -> decltype(auto) {
        return invoke(*parent_->fun_, iters[n]...);
    }, inner_.current_);
```

```
friend constexpr bool operator==(const iterator& x, const iterator& y);
friend constexpr bool operator<(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator>(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator<=(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr bool operator>=(const iterator& x, const iterator& y)
    requires random_access_range<Base>;
friend constexpr auto operator<=>(const iterator& x, const iterator& y)
    requires random_access_range<Base> && three_way_comparable<inner-iterator<Const>>;
```

13 Let *op* be the operator.

14 *Effects:* Equivalent to: `return x.inner_ op y.inner_;`

```
friend constexpr iterator operator+(const iterator& i, difference_type n)
    requires random_access_range<Base>;
friend constexpr iterator operator+(difference_type n, const iterator& i)
    requires random_access_range<Base>;
```

15 *Effects:* Equivalent to: `return iterator(*i.parent_, i.inner_ + n);`

```
friend constexpr iterator operator-(const iterator& i, difference_type n)
    requires random_access_range<Base>;
```

16 *Effects:* Equivalent to: `return iterator(*i.parent_, i.inner_ - n);`

```
friend constexpr difference_type operator-(const iterator& x, const iterator& y)
    requires sized_sentinel_for<inner-iterator<Const>, inner-iterator<Const>>;
```

17 *Effects:* Equivalent to: `return x.inner_ - y.inner_;`

§ 24.7.?.4 Class template `adjacent_transform_view::sentinel` [range.adjacent.transform.sentinel]

```
namespace std::ranges {
    template<forward_range V, copy_constructible F, size_t N>
        requires view<V> && (N > 0) && is_object_v<F> &&
```

```

        regular_invocable<F&, REPEAT(range_reference_t<V>, N)...> &&
        can-reference<invoke_result_t<F&, REPEAT(range_reference_t<V>, N)...>>
template<bool Const>
class adjacent_transform_view<V, F, N>::sentinel {
    inner-sentinel<Const> inner_; // exposition only
    constexpr explicit sentinel(inner-sentinel<Const> inner); // exposition only
public:
    sentinel() = default;
    constexpr sentinel(sentinel<!Const> i)
        requires Const && convertible_to<inner-sentinel<false>, inner-sentinel<Const>>;

    template<bool OtherConst>
        requires sentinel_for<inner-sentinel<Const>, inner-iterator<OtherConst>>
    friend constexpr bool operator==(const iterator<OtherConst>& x, const sentinel&
        y);

    template<bool OtherConst>
        requires sized_sentinel_for<inner-sentinel<Const>, inner-iterator<OtherConst>>
    friend constexpr range_difference_t<maybe-const<OtherConst>, InnerView>>
        operator-(const iterator<OtherConst>& x, const sentinel& y);

    template<bool OtherConst>
        requires sized_sentinel_for<inner-sentinel<Const>, inner-iterator<OtherConst>>
    friend constexpr range_difference_t<maybe-const<OtherConst>, InnerView>>
        operator-(const sentinel& x, const iterator<OtherConst>& y);
};
}

```

```
constexpr explicit sentinel(inner-sentinel<Const> inner);
```

1 *Effects:* Initializes *inner_* with *inner*.

```
constexpr sentinel(sentinel<!Const> i)
    requires Const && convertible_to<inner-sentinel<false>, inner-sentinel<Const>>;
```

2 *Effects:* Initializes *inner_* with `std::move(i.inner_)`.

```
template<bool OtherConst>
    requires sentinel_for<inner-sentinel<Const>, inner-iterator<OtherConst>>
friend constexpr bool operator==(const iterator<OtherConst>& x, const sentinel& y);
```

3 *Effects:* Equivalent to `return x.inner_ == y.inner_;`

```
template<bool OtherConst>
    requires sized_sentinel_for<inner-sentinel<Const>, inner-iterator<OtherConst>>
friend constexpr range_difference_t<maybe-const<OtherConst>, InnerView>>
    operator-(const iterator<OtherConst>& x, const sentinel& y);

template<bool OtherConst>
    requires sized_sentinel_for<inner-sentinel<Const>, inner-iterator<OtherConst>>
friend constexpr range_difference_t<maybe-const<OtherConst>, InnerView>>
    operator-(const sentinel& x, const iterator<OtherConst>& y);
```

4 *Effects:* Equivalent to `return x.inner_ - y.inner_;`

§ 5.10 Feature-test macro

Add the following macro definition to 17.3.2 [\[version.syn\]](#), header `<version>` synopsis, with the value selected by the editor to reflect the date of adoption of this paper:

```
#define __cpp_lib_ranges_zip 20XXXXL // also in <ranges>, <tuple>, <utility>
```

§ 6 Acknowledgements

Thanks to Barry Revzin for implementing this entire paper from spec and finding several wording issues in the process.

§ 7 References

[LWG3526] Casey Carter. Return types of `uses_allocator_construction_args` unspecified.
<https://wg21.link/lwg3526>

[LWG3527] Tim Song. `uses_allocator_construction_args` handles rvalue pairs of rvalue references incorrectly.
<https://wg21.link/lwg3527>

[N4885] Thomas Köppe. 2021-03-17. Working Draft, Standard for Programming Language C++.
<https://wg21.link/n4885>

[P2214R0] Barry Revzin, Conor Hoekstra, Tim Song. 2020-10-15. A Plan for C++23 Ranges.
<https://wg21.link/p2214r0>

[P2325R3] Barry Revzin. 2021-05-14. Views should not be required to be default constructible.
<https://wg21.link/p2325r3>

[range-v3.1592] kitegi. 2020. `zip` does not satisfy the semantic requirements of `bidirectional_iterator` when the ranges have different lengths.
<https://github.com/ericniebler/range-v3/issues/1592>

[range-v3.704] Eric Niebler. 2017. Demand-driven view strength weakening.
<https://github.com/ericniebler/range-v3/issues/704>